

PRODUCT NAME

SIMPONI® Solution for Injection

SIMPONI® IV 12.5mg/1ml Concentrate for Solution for Infusion

DOSAGE FORMS AND STRENGTHS

Golimumab is a human IgG1κ monoclonal antibody that exhibits multiple glycoforms with predicted molecular masses ranging from 149802 daltons to 151064 daltons. Golimumab is produced by a recombinant cell line cultured by continuous perfusion and is purified by a series of steps that includes measures to inactivate and remove viruses.

SIMPONI is available as a solution for injection for subcutaneous (SC) administration in the following presentations:

Autoinjector/Pre-filled pen:

- Each 50 mg single-use autoinjector/pre-filled pen contains 50 mg golimumab per 0.5 mL in an autoinjector/pre-filled pen.
- Each 100 mg single-use autoinjector/pre-filled pen contains 100 mg golimumab per 1.0 mL in an autoinjector/pre-filled pen.

SIMPONI is available as a solution for intravenous (IV) infusion:

Single-use vial

Each single-use vial contains 50 mg golimumab per 4 mL vial (or 12.5 mg of golimumab per mL).

For excipients, see *List of Excipients*.

CLINICAL INFORMATION

Indications

SIMPONI® Solution for Injection

Rheumatoid arthritis (RA):

SIMPONI, by SC administration, in combination with methotrexate (MTX), is indicated for:

- the treatment of moderate to severe, active rheumatoid arthritis in adult patients when the response to disease-modifying anti-rheumatic drug (DMARD) therapy including MTX has been inadequate.
- the treatment of severe, active, and progressive rheumatoid arthritis in adults not previously treated with MTX.

SIMPONI, in combination with MTX, has been shown to reduce the rate of progression of joint damage as measured by X-ray and to improve physical function.

Psoriatic arthritis (PsA):

SIMPONI, by SC administration, alone or in combination with MTX, is indicated for:

- the treatment of active and progressive psoriatic arthritis in adult patients when the response to previous DMARD therapy has been inadequate.

SIMPONI has been shown to reduce the rate of progression of peripheral joint damage as measured by X-ray in patients with polyarticular symmetrical subtypes of the disease and to improve physical function.

Axial spondyloarthritis

Ankylosing spondylitis (AS):

SIMPONI, by SC administration, is indicated for:

- the treatment of severe, active ankylosing spondylitis in adults who have responded inadequately to

conventional therapy.

Non-radiographic axial spondyloarthritis (nr-Axial SpA):

SIMPONI, by SC administration, is indicated for:

- the treatment of adults with severe, active non-radiographic axial spondyloarthritis with objective signs of inflammation as indicated by elevated C-reactive protein (CRP) and/or magnetic resonance imaging (MRI) evidence, who have had an inadequate response to, or are intolerant to nonsteroidal anti-inflammatory drugs (NSAIDs).

Ulcerative colitis (UC):

SIMPONI, by SC administration, is indicated for:

- the treatment of moderately to severely active ulcerative colitis in adult patients who have had an inadequate response to conventional therapy including corticosteroids and 6-mercaptopurine (6-MP) or azathioprine (AZA), or who are intolerant to or have medical contraindications for such therapies.

SIMPONI® IV 12.5mg/1ml Concentrate for Solution for Infusion

Rheumatoid arthritis (RA):

SIMPONI, by IV administration, in combination with MTX, is indicated for:

- the treatment of moderate to severe, active rheumatoid arthritis in adults when the response to disease modifying anti rheumatic drug (DMARD) therapy including MTX has been inadequate.

SIMPONI, in combination with MTX, has been shown to reduce the rate of progression of joint damage as measured by X ray and to improve physical function.

Ankylosing spondylitis (AS):

SIMPONI, by IV administration, is indicated for:

- the treatment of adult patients with active ankylosing spondylitis who have had an inadequate response or intolerance to conventional therapies.

Psoriatic arthritis (PsA):

SIMPONI, by IV administration, alone or in combination with MTX, is indicated for:

- the treatment of active psoriatic arthritis in adult patients when the response to previous DMARD therapy has been inadequate.

Dosage and Administration

Treatment is to be initiated and supervised by qualified physicians experienced in the diagnosis and treatment of rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis, non- radiographic axial spondyloarthritis or ulcerative colitis.

SIMPONI is administered by subcutaneous injection or intravenous infusion.

The efficacy and safety of switching between IV and SC formulations have not been established.

Posology

SIMPONI® Solution for Injection

Rheumatoid arthritis:

SIMPONI 50 mg given once a month, on the same date each month.

SIMPONI should be given concomitantly with MTX.

Psoriatic arthritis, ankylosing spondylitis or non-radiographic axial spondyloarthritis:

SIMPONI 50 mg given once a month, on the same date each month.

For all of the above indications, available data suggest that clinical response is usually achieved within 12 to 14 weeks of treatment (after 3-4 doses). Continued therapy should be reconsidered in patients who show no evidence of therapeutic benefit within this time period.

Patients with body weight greater than 100 kg

For all of the above indications, in patients with RA, PsA, AS, or nr-Axial SpA with a body weight of more than 100 kg who do not achieve an adequate clinical response after 3 or 4 doses, increasing the dose of golimumab to 100 mg once a month may be considered, taking into account the increased risk of certain serious adverse reactions with the 100 mg dose compared with the 50 mg dose (see *Adverse Reactions*). Continued therapy should be reconsidered in patients who show no evidence of therapeutic benefit after receiving 3 to 4 additional doses of 100 mg.

Ulcerative colitis:

Patients with body weight less than 80 kg

SIMPONI given as an initial dose of 200 mg, followed by 100 mg at week 2. Patients who have an adequate response should receive 50 mg at week 6 and every 4 weeks thereafter. Patients who have an inadequate response may benefit from continuing with 100mg at week 6 and every 4 weeks thereafter (see *Pharmacodynamic Properties*).

Patients with body weight greater than or equal to 80 kg

SIMPONI given as an initial dose of 200 mg, followed by 100 mg at week 2, then 100 mg every 4 weeks, thereafter (see *Pharmacodynamic Properties*).

During maintenance treatment, corticosteroids may be tapered in accordance with clinical practice guidelines.

Available data suggest that clinical response is usually achieved within 12-14 weeks of treatment (after 4 doses). Continued therapy should be reconsidered in patients who show no evidence of therapeutic benefit within this time period.

SIMPONI® IV 12.5mg/1ml Concentrate for Solution for Infusion

Administration by intravenous infusion is indicated only for rheumatoid arthritis, ankylosing spondylitis and psoriatic arthritis.

Rheumatoid arthritis, ankylosing spondylitis and psoriatic arthritis

2 mg/kg of SIMPONI given as a 30-minute intravenous infusion at Weeks 0 and 4, then every 8 weeks thereafter (see *Instructions for Use and Handling and Disposal*).

Missed dose

If a patient forgets to inject SIMPONI on the planned date, the forgotten dose should be injected as soon as the patient remembers. Patients should be instructed not to inject a double dose to make up for the forgotten dose.

The next dose should be administered based on the following guidance:

- if the dose is less than 2 weeks late, the patient should inject his/her forgotten dose and stay on his/her original schedule.
- if the dose is more than 2 weeks late, the patient should inject his/her forgotten dose and a new schedule should be established from the date of this injection.

Special populations

Elderly (≥ 65 years)

No dosage adjustment is required in the elderly.

Renal and hepatic impairment

SIMPONI has not been studied in these patient populations. No dose recommendations can be made.

Paediatric patients

The safety and efficacy of SIMPONI in patients aged less than 18 have not yet been established. No data are available.

Method of administration

Subcutaneous injection

SIMPONI® Solution for Injection is for subcutaneous use. After proper training in subcutaneous injection technique, patients may self-inject with SIMPONI® Solution for Injection if their physician determines that this is appropriate, with medical follow-up as necessary. Patients should be instructed to inject the full amount of SIMPONI® Solution for Injection according to the comprehensive instructions for administration provided in the package leaflet. If multiple injections are required, the injections should be administered at different sites on the body.

For administrative instructions, see “Instructions for use, handling and disposal”.

Intravenous infusion

Intravenous infusion of SIMPONI® IV 12.5mg/1ml Concentrate for Solution for Infusion should be administered by qualified health care professionals trained to detect any infusion-related issues.

Comprehensive instructions for the intravenous infusion of SIMPONI® IV 12.5mg/1ml Concentrate for Solution for Infusion are given in “Instructions for use, handling, and disposal.”

Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in *List of Excipients*.

Active tuberculosis (TB) or other severe infections such as sepsis, and opportunistic infections (see *Warnings and Precautions*).

Moderate or severe heart failure (NYHA class III/IV) (see *Warnings and Precautions*).

Warnings and Precautions

Infections

Bacterial (including sepsis and pneumonia), mycobacterial (tuberculosis), invasive fungal and opportunistic infections, including fatalities, have been reported in patients receiving TNF-blocking agents, including SIMPONI. Patients have frequently presented disseminated rather than localized disease. Some of these serious infections have occurred in patients on concomitant immunosuppressive therapy that, in addition to their underlying disease, could predispose them to infections.

For patients who have resided in or traveled to regions where invasive fungal infections such as histoplasmosis, coccidioidomycosis, or blastomycosis are endemic, the benefits and risks of SIMPONI treatment should be carefully considered before initiation or continuation of SIMPONI therapy. In at-risk patients treated with SIMPONI, an invasive fungal infection should be suspected if they develop a serious systemic illness. Antigen and antibody testing may be negative in some patients with active infection. Appropriate empiric antifungal therapy should be considered while a diagnostic workup is being performed. The decision to administer empiric antifungal therapy should be made, if feasible, in consultation with a physician with expertise

in the diagnosis and treatment of invasive fungal infections, and should take into account both the risk for severe fungal infection and the risks of antifungal therapy.

SIMPONI should not be given to patients with a clinically important, active infection. Caution should be exercised when considering the use of SIMPONI in patients with a chronic infection or a history of recurrent infection. Patients should be advised of, and avoid, exposure to potential risk factors for infection as appropriate.

Tuberculosis

Patients should be evaluated for tuberculosis risk factors (including close contact with a person with active tuberculosis) and tested for latent tuberculosis infections prior to treatment with SIMPONI. Treatment of latent tuberculosis infections should be initiated prior to therapy with SIMPONI.

Anti-tuberculosis therapy should be considered prior to initiation of SIMPONI in patients with a past history of latent or active tuberculosis, in whom an adequate course of treatment cannot be confirmed.

Tests for latent tuberculosis may yield false negative results, especially in patients who are immunocompromised or severely ill. Prior to initiating SIMPONI, treatment for latent TB should be considered in patients who have significant risk factors for TB despite a negative test for latent tuberculosis. The decision to initiate anti-tuberculosis therapy in these patients should only be made following consultation with a physician with expertise in the treatment of tuberculosis and taking into account both the risk for latent tuberculosis infection and the risks of anti-tuberculosis therapy.

In patients receiving SIMPONI, tuberculosis has frequently presented as disseminated or extrapulmonary disease. Cases of active tuberculosis have occurred in patients treated with SIMPONI during and after treatment for latent tuberculosis. Patients receiving SIMPONI should be monitored closely for signs and symptoms of active tuberculosis, including patients who tested negative for latent tuberculosis, patients who are on treatment for latent tuberculosis, or patients who were previously treated for tuberculosis infection.

Malignancies

The potential role of TNF-blocking therapy in the development of malignancies is not known. Caution should be exercised when considering TNF-blocking therapy for patients with a history of malignancy or when considering continuing treatment in patients who develop malignancy.

Pediatric malignancy

Postmarketing cases of malignancies, some fatal, have been reported among children, adolescents, and young adults (up to 22 years of age) who received TNF-blocking agents (initiation of therapy $\leq$ 18 years of age) to treat Juvenile Idiopathic Arthritis (JIA), Crohn's disease, or other conditions. Approximately half of the reports were lymphomas. The other cases represented a variety of different malignancies and included malignancies that are not usually observed in children and adolescents. Most of the patients were receiving concomitant immunosuppressants, such as methotrexate, azathioprine, or 6-mercaptopurine. The role of TNF blockers in the development of malignancies in children and adolescents remains unclear.

Lymphoma

In the controlled portions of clinical trials of all the TNF-blocking agents including SIMPONI, more cases of lymphoma have been observed among patients receiving anti-TNF treatment compared with control patients. During the SIMPONI Phase 2 and Phase 3 clinical trials in RA, PsA and AS, the incidence of lymphoma in SIMPONI-treated patients was higher than expected in the general population. Patients with rheumatoid arthritis and other chronic inflammatory diseases, particularly patients with highly active disease and/or chronic exposure to immunosuppressant therapies, may be at higher risk (up to several fold) than the general population for the development of lymphoma, even in the absence of TNF-blocking therapy.

Rare postmarketing cases of hepatosplenic T-cell lymphoma (HSTCL) have been reported in patients treated with other TNF-blocking agents (see *Adverse Reactions*). This rare type of T-cell lymphoma has a very aggressive disease course and is usually fatal. Nearly all of these cases have occurred in patients with Crohn's disease or ulcerative colitis. The majority were in adolescent and young adult males. Almost all these patients had received treatment with azathioprine (AZA) or 6-mercaptopurine (6-MP) concomitantly with a TNF blocker at or prior to diagnosis. The potential risk with the combination of AZA or 6-MP and SIMPONI should be carefully considered. A risk for the development for hepatosplenic T-cell lymphoma in patients treated with TNF blockers cannot be excluded.

Leukemia

Cases of acute and chronic leukemia have been reported with TNF-blocker use, including SIMPONI, in rheumatoid arthritis and other indications. Even in the absence of TNF-blocker therapy, patients with rheumatoid arthritis may be at a higher risk (approximately 2-fold) than the general population for the development of leukemia.

Malignancies other than lymphoma

In the controlled portions of the SIMPONI Phase 2 and Phase 3 clinical trials in RA, PsA, AS, and UC, the incidence of non-lymphoma malignancies (excluding nonmelanoma skin cancer) was similar between the SIMPONI and the control groups.

In an exploratory clinical trial evaluating the use of SIMPONI in patients with severe persistent asthma, more patients treated with SIMPONI reported malignancies compared with control patients (see *Adverse Reactions*). The significance of this finding is unknown.

Colon dysplasia/carcinoma

It is not known if SIMPONI treatment influences the risk for developing dysplasia or colon cancer. All patients with ulcerative colitis who are at increased risk for dysplasia or colon carcinoma (for example, patients with long-standing ulcerative colitis or primary sclerosing cholangitis), or who had a prior history of dysplasia or colon carcinoma, should be screened for dysplasia at regular intervals before therapy and throughout their disease course. This evaluation should include colonoscopy and biopsies per local recommendations. In patients with newly diagnosed dysplasia treated with SIMPONI, the risks and benefits to the individual patient must be carefully reviewed and consideration should be given to whether therapy should be continued.

Skin cancers

Melanoma and Merkel cell carcinoma have been reported in patients treated with TNF-blocking agents, including SIMPONI (see *Adverse Reactions*). Periodic skin examination is recommended for all patients, particularly those with risk factors for skin cancer.

Hepatitis B virus reactivation

As observed with the use of other immunosuppressive drugs, the use of TNF-blocking agents, including SIMPONI, has been associated with reactivation of hepatitis B virus (HBV) in patients who are chronic carriers of the virus (i.e., surface antigen positive). Patients should be tested for HBV infection before initiating treatment with immunosuppressants, including SIMPONI. For patients who test positive for hepatitis B surface antigen, consultation with a physician with expertise in the treatment of hepatitis B is recommended. Chronic carriers of hepatitis B should be appropriately evaluated and monitored prior to the initiation of, during treatment with, and for several months following discontinuation of SIMPONI.

Congestive heart failure (CHF)

Cases of worsening congestive heart failure (CHF) and new onset CHF have been reported with TNF blockers, including SIMPONI. Some cases had a fatal outcome. SIMPONI has not been studied in patients

with CHF. SIMPONI should be used with caution in patients with heart failure. If a decision is made to administer SIMPONI to patients with heart failure, they should be closely monitored during therapy, and SIMPONI should be discontinued if new or worsening symptoms of heart failure appear.

Demyelinating disorders

Use of TNF-blocking agents have been associated with cases of new onset or exacerbation of clinical symptoms and/or radiographic evidence of central nervous system demyelinating disorders, including multiple sclerosis (MS) and peripheral demyelinating disorders, including Guillain-Barré syndrome. Prescribers should exercise caution in considering the use of TNF blockers, including SIMPONI, in patients with central or peripheral nervous system demyelinating disorders. Discontinuation of SIMPONI should be considered if these disorders develop.

Autoimmune processes

Treatment with TNF blockers, including SIMPONI, may result in the formation of antinuclear antibodies (ANA) and, rarely, in the development of a lupus-like syndrome (see *Adverse Reactions*). If a patient develops symptoms suggestive of a lupus-like syndrome following treatment with SIMPONI, treatment should be discontinued.

Concurrent administration of SIMPONI with anakinra

Serious infections and neutropenia were seen in clinical studies with concurrent use of anakinra and another TNF-blocking agent, etanercept, with no added clinical benefit. Because of the nature of the adverse events seen with this combination therapy, similar toxicities may also result from the combination of anakinra and other TNF-blocking agents. Therefore, the combination of SIMPONI and anakinra is not recommended.

Concurrent administration of SIMPONI with abatacept

In clinical studies, concurrent administration of TNF-blocking agents and abatacept have been associated with an increased risk of infections including serious infections compared with TNF-blocking agents alone, without increased clinical benefit. Because of the nature of the adverse events seen with the combination of TNF-blocking agents and abatacept therapy, the combination of SIMPONI and abatacept is not recommended.

Concurrent administration with other biological therapeutics

There is insufficient information regarding the concomitant use of SIMPONI with other biological therapeutics used to treat the same conditions as SIMPONI. The concomitant use of SIMPONI with these biologics is not recommended because of the possibility of an increased risk of infection.

Switching between biological therapeutics

When switching from one biologic to another, patients should continue to be monitored since overlapping biological activity may further increase the risk of infection.

Hematologic reactions

There have been reports of pancytopenia, leukopenia, neutropenia, agranulocytosis, and thrombocytopenia in patients receiving TNF blockers, including SIMPONI. Caution should be exercised in patients treated with SIMPONI who have a current or past history of significant cytopenias.

Live vaccines/therapeutic infectious agents

Patients treated with SIMPONI may receive concurrent vaccinations, except for live vaccines. In patients receiving anti-TNF therapy, limited data are available on the response to vaccination with live vaccines, or on the secondary transmission of infection by live vaccines. Use of live vaccines could result in clinical infections, including disseminated infections.

Other uses of therapeutic infectious agents such as live attenuated bacteria (e.g., BCG bladder instillation for the

treatment of cancer) could result in clinical infections, including disseminated infections. It is recommended that therapeutic infectious agents not be given concurrently with SIMPONI.

Non-live vaccines

Psoriatic arthritis patients treated with SIMPONI in one Phase 3 PsA study were able to mount effective B-cell immune responses to pneumococcal polysaccharide vaccine. Similar numbers of psoriatic arthritis patients receiving SIMPONI and not receiving SIMPONI had at least a 2-fold increase in antibody titers. The proportions of patients with response to pneumococcal vaccine were lower among SIMPONI and control-treated patients receiving MTX, compared with patients not receiving MTX. Overall, the data indicates that SIMPONI does not suppress the humoral immune response to this vaccine.

Allergic reactions

Latex sensitivity

The needle cover on the pre-filled syringe in the autoinjector/pre-filled pen contains dry natural rubber (a derivative of latex), which may cause allergic reactions in individuals sensitive to latex.

Hypersensitivity reactions

In postmarketing experience, serious systemic hypersensitivity reactions (including anaphylactic reaction) have been reported following SIMPONI administration. Some of these reactions occurred after the first administration of SIMPONI. If an anaphylactic or other serious allergic reaction occurs, administration of SIMPONI should be discontinued immediately and appropriate therapy instituted.

Special populations

Geriatric use

In the Phase 3 SC studies in RA, PsA, and AS and the Phase 3 IV studies in RA, no overall differences in AEs, SAEs, and serious infections in patients age 65 or older who received SIMPONI were observed, compared with younger patients. In the Phase 3 IV studies in PsA and AS, there were insufficient numbers of patients aged 65 years and over to determine whether they respond differently from patients aged 18 to 65 years old. In UC, there were insufficient numbers of patients aged 65 and over to determine whether they respond differently from patients aged 18 to 65. Because there is a higher incidence of infections in the elderly population in general, caution should be used in treating the elderly. There were no patients aged 65 and over in the nr-Axial SpA study.

Interactions

Specific drug interaction studies have not been conducted with SIMPONI.

Concurrent use of SIMPONI with other biological therapeutics

The combination of SIMPONI with other biological therapeutics used to treat the same conditions as SIMPONI, including anakinra and abatacept, is not recommended (see *Warnings and Precautions*).

Live vaccines/therapeutic infectious agents

Live vaccines should not be given concurrently with SIMPONI (see *Warnings and Precautions*).

Therapeutic infectious agents should not be given concurrently with SIMPONI (see *Warnings and Precautions*).

Methotrexate

No significant effect of methotrexate on the clearance of SIMPONI administered intravenously was observed. Following subcutaneous administration, concomitant use of methotrexate resulted in higher steady-state trough concentrations of SIMPONI in patients with RA, PsA, or AS. However, the data does not suggest the need for dose adjustment of either SIMPONI or methotrexate (see *Pharmacokinetic*

Properties).

Pregnancy, Breast-feeding and Fertility

Pregnancy

An embryofetal developmental toxicology study was performed in which pregnant cynomolgus monkeys were treated with golimumab during the first trimester at dosages up to 50 mg/kg twice weekly (over 500-fold higher in terms of dose/body weight ratio than the proposed clinical dose of 50 mg every 4 weeks). The mean peak maternal serum concentration obtained in this study (1576 mcg/mL) is over 900-fold higher than median steady-state $C_{\max}$ value (1.71 mcg/mL) following 50 mg every 4-week SC dosing in patients with RA, PsA, and AS. Umbilical cord blood samples collected at the end of the second trimester showed that fetuses were exposed to golimumab during gestation. Fetal serum concentrations were approximately 50% of the maternal serum concentrations. In this study, *in utero* exposure to golimumab produced no developmental defects to the fetus.

A pre- and postnatal developmental study was performed in which pregnant cynomolgus monkeys were treated with golimumab during the second and third trimesters and during lactation. Golimumab was present in the neonatal serum from the time of birth and for up to 6 months postpartum. The mean peak maternal serum concentration obtained in this study (1482 mcg/mL) is over 860-fold higher than median steady-state $C_{\max}$ value (1.71 mcg/mL) following 50 mg every 4-week SC dosing in patients with RA, PsA, and AS. Exposure to golimumab during gestation and during the postnatal period caused no developmental defects in the infants. However, animal reproductive and developmental studies are not always predictive of human response.

Golimumab crosses the placenta. Following treatment with another TNF-blocking monoclonal antibody during pregnancy, the antibody has been detected for up to 6 months in the serum of the infant born by the treated women. Consequently, these infants may be at increased risk of infection. Administration of live vaccines to infants exposed to golimumab *in utero* is not recommended for 6 months following the mother's last golimumab injection during pregnancy (see *Warnings and Precautions* and *Interactions*).

There is a moderate amount of data on the use of golimumab in pregnant women. In a prospective population-based study from Northern Europe including 131 pregnancies exposed to SIMPONI, there was no increased risk of major congenital anomaly following in utero exposure to SIMPONI vs non-biologic systemic therapy (adjusted OR 0.79 [95% CI 0.35-1.81] or vs the general population (adjusted OR 0.95 [95% CI 0.42-2.16]).

It is not known whether SIMPONI can cause fetal harm when administered to a pregnant woman or can affect reproduction capacity. SIMPONI should be given to a pregnant woman only if clearly needed.

Breast-feeding

In the pre- and postnatal development study in cynomolgus monkeys (see *Pregnancy* above) in which golimumab was administered during pregnancy and lactation, golimumab was detected in the breast milk at concentrations that were approximately 350-fold lower than the maternal serum concentrations. It is not known whether golimumab is excreted in human milk or absorbed systemically after ingestion. Because many drugs and immunoglobulins are excreted in human milk, and because of the potential for adverse reactions in nursing infants from SIMPONI, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

Effects on Ability to Drive and Use Machines

No studies on the effects on the ability to drive and use machines have been performed.

Adverse Reactions

Throughout this section, adverse reactions are presented. Adverse reactions are adverse events that were considered to be reasonably associated with the use of golimumab based on the comprehensive assessment of the available adverse event information. A causal relationship with golimumab cannot be reliably established in individual cases. Further, because clinical trials are conducted under widely varying conditions, adverse reaction rates observed in the clinical trials of a drug cannot be directly compared to rates in the clinical trials of another drug and may not reflect the rates observed in clinical practice.

Safety data from Phase 2 and 3 clinical trials are available from 6161 golimumab-treated patients including 3090 with rheumatoid arthritis, 634 with PsA, 768 with AS, 1245 with ulcerative colitis, 231 with severe persistent asthma, and 193 with active non-radiographic axial spondyloarthritis (nr-Axial SpA). Adverse reactions observed with golimumab are summarized in Table 1.

In general, the overall safety profile was similar for patients receiving golimumab via the SC or IV routes of administration.

Within the designated system organ classes, the adverse reactions are listed under headings of frequency, using the following convention:

Very common	($\geq 1/10$)
Common	(frequent) ($\geq 1/100$, $< 1/10$)
Uncommon	(infrequent) ($\geq 1/1000$, $< 1/100$)
Rare	($\geq 1/10000$, $< 1/1000$, including isolated reports)
Not known	(cannot be estimated from the available data)

Table 1: Summary of golimumab adverse reactions in clinical studies	
Infections and infestations	
Very common:	Upper respiratory tract infection (nasopharyngitis, pharyngitis, laryngitis, and rhinitis)
Common:	Bacterial infections (such as cellulitis), lower respiratory tract infection (pneumonia), viral infections (such as influenza and herpes), bronchitis, sinusitis, superficial fungal infections, abscess
Uncommon:	Sepsis including septic shock, pyelonephritis
Rare:	Histoplasmosis, coccidioidomycosis, tuberculosis, opportunistic infections (invasive fungal infections, bacterial, atypical mycobacterial and protozoal), hepatitis B reactivation, pneumocystosis, arthritis bacterial, bursitis infective
Neoplasm benign and malignant	
Rare:	Lymphoma, leukemia
Not known:	Pediatric malignancy
Investigations	
Common:	Alanine aminotransferase increased, aspartate aminotransferase increased
Uncommon:	Neutrophil count decreased
Blood and lymphatic system disorders	
Common:	Leukopenia (including neutropenia), anemia
Uncommon:	Thrombocytopenia, pancytopenia
Immune system disorders	

Common:	Autoantibody positive, non-serious allergic reactions
Nervous system disorders	
Common:	Dizziness, paresthesia
Rare:	Demyelinating disorders (central and peripheral)
Cardiac disorders	
Rare:	Congestive heart failure (new onset or worsening)
Vascular disorders	
Common:	Hypertension
Rare:	Vasculitis (systemic)
Respiratory, thoracic and mediastinal disorders	
Uncommon:	Interstitial lung disease
Gastrointestinal disorders	
Uncommon:	Constipation
Skin and subcutaneous tissue disorders	
Common:	Rash, alopecia
Uncommon:	Psoriasis: new onset or worsening, palmar/plantar, and pustular
Rare:	Vasculitis (cutaneous)
Musculoskeletal and connective tissue disorders	
Rare:	Lupus-like syndrome
General disorders and administration site conditions	
Common:	Pyrexia, injection site reaction (injection site erythema, urticaria, induration, pain, bruising, pruritus, irritation, parasthesia)

The data described below reflect adverse reactions from the SC Phase 2 and 3 clinical trials, except for administration reactions and liver enzyme elevations, which include both SC and IV data. Throughout this section, median duration of follow-up (approximately 4 years) is generally presented for all golimumab use. Where golimumab use is described by dose, the median duration of follow-up varies (approximately 2 years for 50 mg dose, approximately 3 years for 100 mg dose) as patients may have switched between doses.

Infections (see *Warnings and Precautions*)

In the controlled period of pivotal trials, upper respiratory tract infection was the most common adverse reaction reported in 12.6% of golimumab-treated patients (incidence per patient-year: 0.61; 95% CI: 0.55, 0.67) compared with 11.0% of control patients (incidence per patient-year: 0.55; 95% CI: 0.46, 0.64). In controlled and uncontrolled portions of the studies with a median follow-up of approximately 4 years, the incidence per patient-year of upper respiratory tract infections was 0.35 events; 95% CI: 0.34, 0.36, for golimumab-treated patients.

In the controlled period of pivotal trials, infections were observed in 23.0% of golimumab-treated patients (incidence per patient-year: 1.32; 95% CI: 1.23, 1.41) compared with 20.2% of control patients (incidence per patient-year: 1.22; 95% CI: 1.09, 1.36). In controlled and uncontrolled portions of the trials with a median follow-up of approximately 4 years, the incidence per patient-year of infections was 0.81 events; 95% CI: 0.79, 0.83, for golimumab-treated patients.

Serious infections observed in golimumab-treated patients included sepsis, pneumonia, cellulitis, abscess, opportunistic infections, and tuberculosis. In the controlled period of RA, PsA, ulcerative colitis, AS, and nr-Axial SpA trials, serious infections were observed in 1.2% of golimumab-treated patients

and 1.2% of control-treated patients. The incidence of serious infections per patient-year of follow-up in the controlled period of RA, PsA, AS, and nr-Axial SpA trials was 0.07; 95% CI: 0.05, 0.11 for the golimumab 100-mg group, 0.03; 95% CI: 0.01, 0.06 for the golimumab 50 mg-group and 0.04; 95% CI: 0.02, 0.07 for the placebo group. In the controlled period of UC trials of golimumab induction, serious infections were observed in 0.8% of golimumab-treated patients compared with 1.5% of control-treated patients. In the controlled and uncontrolled portions of the pivotal trials with a median follow-up of up to 3 years, there was a greater incidence of serious infections, including opportunistic infections and TB in patients receiving golimumab 100 mg, when compared with patients receiving golimumab 50 mg. The incidence per patient-year of all serious infections was 0.04; 95% CI: 0.04, 0.05, in patients receiving golimumab 100 mg and 0.03; 95% CI: 0.02, 0.03, in patients receiving golimumab 50 mg. These results may be confounded by the designs of the pivotal trials and different durations of follow-up across treatment groups.

Malignancies (see *Warnings and Precautions*)

Lymphoma

The incidence of lymphoma in golimumab-treated patients during the pivotal trials was higher than expected in the general population. In the controlled and uncontrolled portions of these trials with a median follow-up of up to 3 years, a greater incidence of lymphoma was observed in patients receiving golimumab 100 mg compared with patients receiving golimumab 50 mg. These results may be confounded by the small number of events, designs of the Phase 3 studies, and different durations of follow-up across treatment groups. The majority of lymphomas occurred in RA Study 2, which enrolled patients previously exposed to anti-TNF agents, who had longer disease duration and more refractory disease.

Malignancies other than lymphoma

In the controlled periods of pivotal trials, the incidence of non-lymphoma malignancies (excluding nonmelanoma skin cancer) was similar between the golimumab and the control groups. Through approximately 4 years of follow-up, the incidence of non-lymphoma malignancies (excluding non-melanoma skin cancer) was similar to the general population.

In an exploratory clinical trial involving patients with severe persistent asthma, more patients treated with golimumab had malignancies compared with control patients. The significance of this finding in the asthma population is unknown.

The potential role of TNF-blocking therapy in the development of malignancies is unknown.

Demyelinating disorders (see *Warnings and Precautions*)

In the controlled and uncontrolled periods of the pivotal trials with a median follow-up of up to 3 years, a greater incidence of demyelination was observed in patients receiving golimumab 100 mg compared with patients receiving golimumab 50 mg. These results may be confounded by the small number of events, designs of the pivotal trials, and different durations of follow-up across treatment groups.

Liver enzyme elevations

In the controlled period of RA and PsA pivotal trials, mild ALT elevations (> 1 and $< 3 \times$ ULN) occurred in similar proportions of golimumab-treated and control patients (22.1% to 27.4% of patients); in the AS and nr-Axial SpA studies, more golimumab-treated patients (26.9%) than control patients (10.6%) had mild ALT elevations. In the controlled and uncontrolled periods of the RA and PsA pivotal trials, with a median follow-up of approximately 5 years, the incidence of mild ALT elevations was similar in golimumab-treated and control patients.

In the controlled period of the UC pivotal trials of golimumab induction, mild ALT elevations (> 1 and $< 3 \times$ ULN) occurred in similar proportions of golimumab-treated and control patients (7.8% to 6.9%,

respectively). In controlled and uncontrolled periods of the UC pivotal trials with a median follow-up of approximately 2 years, the proportion of patients with mild ALT elevations was 24.7% in patients receiving golimumab.

In the controlled period of RA and AS pivotal trials, ALT elevations $\geq 5 \times \text{ULN}$ were uncommon and seen in more golimumab-treated patients (0.4% to 0.9%) than control patients (0.0%). This trend was not observed in the PsA population. In the controlled and uncontrolled periods of RA, PsA, and AS pivotal trials, with a median follow-up of 5 years, the incidence of ALT elevations $\geq 5 \times \text{ULN}$ was similar in both golimumab-treated and control patients. The majority of these elevations were asymptomatic. No cases were reported in the controlled and uncontrolled periods of the nr-Axial SpA study (up to 1 year).

In the controlled periods of the pivotal UC trials of golimumab induction, ALT elevations $\geq 5 \times \text{ULN}$ occurred in similar proportions of golimumab-treated patients compared to placebo-treated patients (0.3% to 1.0%, respectively). In the controlled and uncontrolled periods of the pivotal UC trials with a median follow-up of approximately 2 years, the proportion of patients with ALT elevations $\geq 5 \times \text{ULN}$ was 0.8% in patients receiving golimumab.

In the IV pivotal trials, liver enzyme elevations were comparable with those observed in the SC studies with the exception of the following:

In the controlled period of the IV PsA pivotal trial, mild ALT elevations (> 1 and $< 3 \times \text{ULN}$) were observed in more golimumab-treated patients (34%) than control patients (26%).

In the controlled period of the IV PsA pivotal trial, ALT elevations ≥ 3 and $< 5 \times \text{ULN}$ were observed in more golimumab-treated patients (2.9%) than control patients (0.4%).

In the controlled period of the IV PsA pivotal trial, ALT elevations $\geq 5 \times \text{ULN}$ were observed in more golimumab-treated patients (1.7%) than control patients (0.4%).

Administration reactions

In the controlled periods of the pivotal trials, 5.4% of golimumab-treated patients had injection site reactions compared with 2.0% in control-treated patients. The majority of the injection site reactions were mild and moderate, and the most frequent manifestation was injection site erythema.

In the controlled periods of the pivotal IV trials, 0.2% of placebo-treated subjects and 2.8% of golimumab-treated subjects had an infusion reaction. The most common infusion reactions were rash and headache. No serious infusion reactions were reported.

In controlled Phase 2 and/or 3 trials in RA, PsA, AS, nr-Axial SpA, severe persistent asthma, and Phase 2/3 trials in UC, no patients treated with golimumab developed anaphylactic reactions deemed to be related to golimumab.

Antinuclear antibodies (ANA)/anti-double-stranded DNA (dsDNA) antibodies

In the controlled and uncontrolled periods of the pivotal trials through 1 year of follow-up, 3.5% of golimumab-treated patients and 2.3% of control patients were newly ANA-positive (at titers of 1:160 or greater). The frequency of anti-dsDNA antibodies at 1 year of follow-up in patients anti-dsDNA negative at baseline was 1.1% (see *Warnings and Precautions*).

Postmarketing experience

The frequencies provided below reflect reporting rates of adverse reactions from the worldwide postmarketing experience with golimumab. Precise estimates of incidence cannot be made due to voluntary reporting from a population of uncertain size. These adverse reactions are ranked by frequency, using the

following convention: Very common ($\geq 1/10$), Common ($\geq 1/100$ and $< 1/10$), Uncommon ($\geq 1/1000$ and $< 1/100$), Rare ($\geq 1/10000$ and $< 1/1000$), Very rare ($< 1/10000$, including isolated reports), and Not known (cannot be estimated from the available data).

Table 2: Golimumab postmarketing adverse reactions

System Organ Class	Adverse Reaction	Frequency
General Disorders and Administration Site Conditions	Infusion-related reaction	Common
Neoplasm benign and malignant	Melanoma Merkel cell carcinoma Hepatosplenic T-cell lymphoma* Kaposi's sarcoma	Rare Not known
Immune system disorders	Serious systemic hypersensitivity reactions (including anaphylactic reaction) Sarcoidosis	Rare
Skin and subcutaneous tissue disorders	Bullous skin reactions Lichenoid reactions Skin exfoliation	Uncommon Rare

* Observed with other TNF-blocking agents

Overdose

Single doses up to 10 mg/kg intravenously have been administered in a clinical study without dose-limiting toxicity. In case of overdosage, it is recommended that the patient be monitored for any signs or symptoms of adverse effects and appropriate symptomatic treatment be instituted immediately.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic Properties

Pharmacotherapeutic group: Immunosuppressants, tumour necrosis factor alpha (TNF- α) inhibitors, ATC code: L04AB06

Mechanism of action

Golimumab is a human monoclonal antibody that forms high affinity, stable complexes with both the soluble and transmembrane bioactive forms of human TNF, which prevents the binding of TNF- α to its receptors.

Pharmacodynamic effects

The binding of human TNF by golimumab was shown to neutralise TNF- α -induced cell-surface expression of the adhesion molecules E-selectin, vascular cell adhesion molecule (VCAM)-1 and intercellular adhesion molecule (ICAM)-1 by human endothelial cells. *In-vitro*, TNF-induced secretion of interleukin (IL)-6, IL-8 and granulocyte-macrophage colony stimulating factor (GM-CSF) by human endothelial cells was also inhibited by golimumab.

Improvement in C-reactive protein (CRP) levels were observed relative to placebo groups and treatment with SIMPONI resulted in significant reductions from baseline in serum levels of IL-6, ICAM-1, matrix metalloproteinase-3 (MMP-3) and vascular endothelial growth factor (VEGF) compared to control treatment. In addition, levels of TNF- α were reduced in RA and AS patients and levels of IL-8 were reduced in PsA patients. These changes were observed at the first assessment (week 4) after the initial SIMPONI administration and were generally maintained through week 24.

Clinical efficacy

Rheumatoid arthritis:

The efficacy of SIMPONI was demonstrated in three multi-centre, randomised, double-blind, placebo-controlled studies in over 1500 patients ≥ 18 years of age with moderately to severely active RA diagnosed according to American College of Rheumatology (ACR) criteria for at least 3 months prior to screening. Patients had at least 4 swollen and 4 tender joints. SIMPONI or placebo were subcutaneously administered every 4 weeks.

GO-FORWARD evaluated 444 patients who had active RA despite a stable dose of at least 15 mg/week of MTX and who had not been previously treated with an anti-TNF agent. Patients were randomised to receive placebo + MTX, SIMPONI 50 mg + MTX, SIMPONI 100 mg + MTX or SIMPONI 100 mg + placebo. Patients receiving placebo + MTX were switched to SIMPONI 50 mg + MTX after week 24. At week 52, patients entered an open label long-term extension.

GO-AFTER evaluated 445 patients who were previously treated with one or more of the anti-TNF agents adalimumab, etanercept, or infliximab. Patients were randomised to receive placebo, SIMPONI 50 mg, or SIMPONI 100 mg. Patients were allowed to continue concomitant DMARD therapy with MTX, sulfasalazine (SSZ), and/or hydroxychloroquine (HCQ) during the study. The stated reasons for discontinuation of prior anti-TNF therapies were lack of efficacy (58%), intolerance (13%), and/or reasons other than safety or efficacy (29%, mostly for financial reasons).

GO-BEFORE evaluated 637 patients with active RA who were MTX-naïve and had not previously been treated with an anti-TNF agent. Patients were randomised to receive placebo + MTX, SIMPONI 50 mg + MTX, SIMPONI 100 mg + MTX or SIMPONI 100 mg + placebo. At week 52, patients entered an open label long-term extension in which patients receiving placebo + MTX who had at least 1 tender or swollen joint were switched to SIMPONI 50 mg + MTX.

In GO-FORWARD, the (co-)primary endpoints were the percentage of patients achieving an ACR 20 response at Week 14 and the improvement from baseline in Health Assessment Questionnaire (HAQ) at Week 24. In GO-AFTER, the primary endpoint was the percentage of patients achieving an ACR 20 response at week 14. In GO-BEFORE, the co-primary endpoints were the percentage of patients achieving ACR 50 response at Week 24 and the change from baseline in the van der Heijde-modified Sharp (vdH-S) score at week 52. In addition to the primary endpoint(s), additional assessments of the impact of SIMPONI treatment on the signs and symptoms of arthritis, radiographic response, physical function and health-related quality of life were performed.

In general, no clinically meaningful differences in measures of efficacy were observed between the SIMPONI 50 mg and 100 mg dosing regimens with concomitant MTX, through week 104 in GO-FORWARD and GO-BEFORE and through week 24 in GO-AFTER. In each of the RA studies by study design, patients in the long-term extension may have switched between the 50 mg and 100 mg SIMPONI doses at the discretion of the study physician.

IV Study RA-1 (GO-FURTHER) evaluated 592 patients with active RA despite concurrent MTX therapy. Patients were randomized to receive either SIMPONI 2 mg/kg IV (N=395) or IV placebo (saline) (N=197) at Week 0, Week 4, and every 8 weeks thereafter in addition to their weekly maintenance MTX dose. All patients receiving IV placebo + MTX received SIMPONI 2 mg/kg IV + MTX after Week 24, but the trial remained double-blind through 52 weeks of treatment. The primary endpoint was the percentage of patients achieving ACR 20 response at Week 14. The major secondary endpoints included DAS28 response (using CRP) and change from baseline in HAQ-DI at Week 14 as well as ACR 50 response and change from baseline in vdH-s score at Week 24. Other pre-specified endpoints included improvement in ACR components, ACR response over time, improvement in physical function and health-related quality of life, as well as health economics measures.

Signs and symptoms

Key ACR results for the SIMPONI 50 mg dose at weeks 14, 24 and 52 for GO-FORWARD, GO-AFTER and GO-BEFORE are shown in Table 3 and are described below. Responses were observed at the first assessment (Week 4) after the initial SIMPONI administration.

In GO-FORWARD, among 89 subjects randomised to SIMPONI 50 mg + MTX, 48 were still on this treatment at week 104. Among those, 40, 33 and 24 patients had ACR 20/50/70 response, respectively at week 104. Among patients remaining in the study and treated with SIMPONI, similar rates of ACR 20/50/70 response was observed from week 104 through week 256.

In GO-AFTER, the percentage of patients achieving an ACR 20 response was greater for patients receiving SIMPONI than for patients receiving placebo regardless of the reason reported for discontinuation of one or more prior anti-TNF therapies.

Table 3: Key efficacy outcomes from the controlled portions of GO-FORWARD, GO-AFTER and GO-BEFORE

	GO-FORWARD Active RA despite MTX		GO-AFTER Active RA, previously treated with one or more anti-TNF agent(s)		GO-BEFORE Active RA, MTX Naïve	
	Placebo + MTX	SIMPONI 50 mg + MTX	Placebo	SIMPONI 50 mg	Placebo + MTX	SIMPONI 50 mg + MTX
n ^a	133	89	150	147	160	159
Responders, % of patients						
ACR 20						
Week 14	33%	55%*	18%	35%*	NA	NA
Week 24	28%	60%*	16%	31% p=0.002	49%	62%
Week 52	NA	NA	NA	NA	52%	60%
ACR 50						
Week 14	10%	35%*	7%	15% p=0.021	NA	NA
Week 24	14%	37%*	4%	16%*	29%	40%
Week 52	NA	NA	NA	NA	36%	42%
ACR 70						
Week 14	4%	14% p=0.008	2%	10% p=0.005	NA	NA
Week 24	5%	20%*	2%	9% p=0.009	16%	24%
Week 52	NA	NA	NA	NA	22%	28%
a: n reflects randomised patients; actual number of patients evaluable for each endpoint may vary by timepoint.						
*: p ≤ 0.001						
NA: Not applicable						

In GO-BEFORE the primary analysis in patients with moderate to severe rheumatoid arthritis (combined SIMPONI 50 and 100 mg + MTX groups vs MTX alone for ACR50) was not statistically significant at week 24 (p = 0.053). At week 52 in the overall population, the percentage of patients in the SIMPONI 50 mg + MTX group who achieved an ACR response was generally higher but not significantly different when compared with MTX alone (see Table 3). Additional analyses were performed in subsets representative of the indicated population of patients with severe, active and progressive RA. A generally greater effect of SIMPONI 50 mg + MTX versus MTX alone was demonstrated in the indicated population compared with

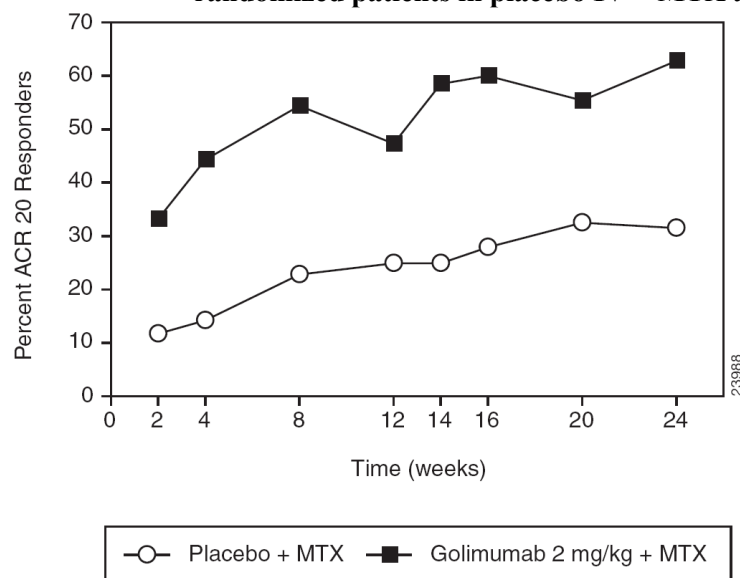
the overall population.

In GO-FORWARD and GO-AFTER, clinically meaningful and statistically significant responses in Disease Activity Scale (DAS)28 were observed at each prespecified time point, at week 14 and at week 24 ($p \leq 0.001$). Among patients who remained on the SIMPONI treatment to which they were randomised at study start, DAS28 responses were maintained through week 104. Among patients remaining in the study and treated with SIMPONI, DAS28 responses were similar from week 104 through week 256.

In GO-BEFORE, major clinical response, defined as the maintenance of an ACR 70 response over a continuous 6-month period, was measured. At week 52, 15% of patients in the SIMPONI 50 mg + MTX group achieved a major clinical response compared with 7% of patients in the placebo + MTX group ($p = 0.018$). Among 159 subjects randomised to SIMPONI 50 mg + MTX, 96 were still on this treatment at week 104. Among those, 85, 66 and 53 patients had ACR 20/50/70 response, respectively, at week 104. Among patients remaining in the study and treated with SIMPONI, similar rates of ACR 20/50/70 response were observed from week 104 through week 256.

Treatment with SIMPONI IV in patients with active RA despite MTX resulted in improvement in signs and symptoms as demonstrated by the percentage of patients achieving an ACR 20 response at Week 14. A significantly greater percentage of patients achieved an ACR 20 response in the SIMPONI IV + MTX group than in the placebo IV + MTX group ($p < 0.001$). In the SIMPONI IV + MTX group, 58.5% of patients achieved an ACR 20 response compared with 24.9% in the placebo IV + MTX group at Week 14. The percentage of patients achieving an ACR 20 response at Week 24 was also significantly greater for patients receiving SIMPONI IV + MTX as compared with placebo IV + MTX (62.8% compared with 31.5%, respectively) (Table 4).

Figure 1: IV RA Study 1: Percentage of patients achieving ACR 20 response through Week 24; randomized patients in placebo IV + MTX and SIMPONI IV + MTX dose groups



The percentage of patients achieving ACR 50 and ACR 70 responses was also greater in the SIMPONI IV + MTX group than in the placebo IV + MTX group (Table 4). The percentage of patients achieving an ACR 50 response for the SIMPONI IV + MTX and placebo IV + MTX groups, was 29.9% and 8.6%, respectively, at Week 14 ($p < 0.001$), and 34.9% and 13.2%, respectively, at Week 24 ($p < 0.001$). The percentage of patients achieving an ACR 70 response for the SIMPONI IV + MTX and placebo IV + MTX groups was 12.4% and 3.0%, respectively, at Week 14 ($p < 0.001$) and 17.5% and 4.1%, respectively at

Week 24 ($p<0.001$).

The proportions of patients achieving an ACR 20, ACR 50, or ACR 70 response were maintained after Week 24 through Week 52.

The percentage of patients achieving a DAS28 (using CRP) response was significantly greater for those patients treated with SIMPONI IV + MTX compared with those who received placebo IV + MTX at Week 14 (81.3% compared with 40.1%; $p<0.001$) and at Week 24 (81.0% compared with 44.7%; $p<0.001$). The percentage of patients achieving DAS28 < 2.6 (using CRP) remission was significantly greater for those patients treated with SIMPONI IV + MTX compared with those who received placebo IV + MTX at Week 14 (15.4% compared with 4.6%; $p<0.001$) and at Week 24 (17.7% compared with 5.1%; $p<0.001$). Of patients treated with SIMPONI + MTX who achieved DAS28 remission, 43% had no active joints, 27% had one active joint, 23% had two active joints, and 7% had three or more active joints, where an active joint was a joint that was rated as tender or swollen or both. The proportions of SIMPONI -treated patients achieving a DAS28 response or remission (using CRP) at Week 24 were maintained through Week 52.

Table 4: IV RA Study 1: Percent of RA patients with ACR responses and DAS responses and remission at Week 14, Week 24, and Week 52

	Placebo IV + MTX (N=197) ^a	SIMPONI IV + MTX (N=395) ^a	p-value
ACR 20 (% responders)			
Week 14	24.9%	58.5%	<0.001
Week 24	31.5%	62.8%	<0.001
Week 52	N/A ^b	65.8%	
ACR 50 (% responders)			
Week 14	8.6%	29.9%	<0.001
Week 24	13.2%	34.9%	<0.001
Week 52	N/A ^b	38.7%	
ACR 70 (% responders)			
Week 14	3.0%	12.4%	<0.001
Week 24	4.1%	17.5%	<0.001
Week 52	N/A ^b	18.2%	
DAS28 (using CRP) response (% responders)			
Week 14	40.1%	81.3%	<0.001
Week 24	44.7%	81.0%	<0.001
Week 52	N/A ^b	81.3%	
DAS28 (using CRP) remission (% remitters)			
Week 14	4.6%	15.4%	<0.001
Week 24	5.1%	17.7%	<0.001
Week 52	N/A ^b	20.3%	

Table 4: IV RA Study 1: Percent of RA patients with ACR responses and DAS responses and remission at Week 14, Week 24, and Week 52

	Placebo IV + MTX (N=197) ^a	SIMPONI IV + MTX (N=395) ^a	p-value
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^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

^b These results are not reported as all patients received SIMPONI + MTX after Week 24 due to study design.

An ACR 20 response (Felson et al, 1995) was defined as:

1. $\geq 20\%$ improvement in swollen joint count (66 joints) and tender joint count (68 joints); and
2. $\geq 20\%$ improvement in 3 of the following 5 assessments:
 - Patient's assessment of pain on a 0-10 cm VAS scale (no pain to the worst possible pain).
 - Patient's global assessment of disease activity on a 0-10 cm VAS scale (very well to very poor).
 - Physician's global assessment of disease activity on a 0-10 cm VAS scale (no active arthritis to extremely active arthritis).
 - Patient's assessment of physical function as measured by the HAQ on a scale of 0 to 3 (without any difficulty to unable to do).
 - CRP.

An ACR 50 or ACR 70 response was defined as $\geq 50\%$ or $\geq 70\%$ improvement in 1 and 2 above.

DAS28 (using CRP) was derived using 28 joint assessments for swelling (swollen 28) and pain/tenderness (tender 28) counts, patient's global assessment of disease activity (GH) using the formula, combined with CRP or ESR: $\text{DAS28} = 0.56(\text{sqrt}(\text{tender } 28)) + 0.28(\text{sqrt}(\text{swollen } 28)) + 0.36(\text{Ln}(\text{CRP}+1)) [\text{or } 0.70(\text{Ln}(\text{ESR}))] + 0.014(\text{GH}) + 0.96$ (van der Linden, 2004; van Riel, 2000). DAS28 response at a visit was based on the current DAS28 score as well as the improvement from baseline in the DAS28 score. Patients were classified as:

Improvement in DAS28 from Baseline

Current DAS28	≥ 1.2	≥ 0.6 to ≤ 1.2	≤ 0.6
≤ 3.2	Good	Moderate	None
> 3.2 to ≤ 5.1	Moderate	Moderate	None
> 5.1	Moderate	None	None

DAS28 responders include patients with moderate or good response.

DAS28 remitters include patients with a DAS28 value of < 2.6 at a visit.

SIMPONI IV + MTX treatment also resulted in significantly greater improvement for each ACR component compared with treatment with placebo + MTX (Table 5). Swollen joint count for the SIMPONI IV + MTX and placebo IV + MTX group improved by 68% and 25%, respectively, at Week 14 and 75% and 27%, respectively, at Week 24. Improvement in tender joint count was 62% compared with 13% at Week 14, and 64% compared with 14% at Week 24, for the SIMPONI IV + MTX and placebo IV + MTX group, respectively. Patients' and physicians' assessments and HAQ score also were significantly improved for SIMPONI IV + MTX compared with placebo IV + MTX at Week 14 and Week 24. For SIMPONI IV + MTX, there was a 78% improvement in CRP compared with 29% improvement for placebo + MTX at Week 14, and 77% compared with 21% at Week 24. The improvement in all ACR components observed at Week 24 was maintained at each visit through Week 52.

Table 5: IV RA Study 1: Percent improvement in ACR components at Week 14, Week 24, and Week 52; randomized patients

	Placebo + MTX (N=197)^a	IV SIMPONI IV + MTX (N=395)^a	p-value
Number of swollen joints			
Baseline (median)	12.0	12.0	
Week 14	25%	68%	p≤0.001
Week 24	27%	75%	p≤0.001
Week 52	N/A ^b	86%	
Number of tender joints			
Baseline (median)	22.0	24.0	
Week 14	13%	62%	p≤0.001
Week 24	14%	64%	p≤0.001
Week 52	N/A ^b	71%	
Patient's assessment of pain			
Baseline (median)	6.9	6.5	
Week 14	7%	38%	p≤0.001
Week 24	11%	43%	p≤0.001
Week 52	N/A ^b	41%	
Patient's global assessment of disease activity			
Baseline (median)	6.9	6.6	
Week 14	7%	37%	p≤0.001
Week 24	14%	41%	p≤0.001
Week 52	N/A ^b	42%	
Physician's global assessment of disease activity			
Baseline (median)	6.3	6.3	
Week 14	15%	52%	p≤0.001
Week 24	24%	55%	p≤0.001
Week 52	N/A ^b	61%	
HAQ score			
Baseline (median)	1.6	1.6	
Week 14	7%	29%	p≤0.001
Week 24	9%	30%	p≤0.001
Week 52	N/A ^b	30%	
CRP (mg/dL)			
Baseline (median)	1.7	2.0	
Week 14	29%	78%	p≤0.001
Week 24	21%	77%	p≤0.001
Week 52	N/A ^b	66%	

Table 5: IV RA Study 1: Percent improvement in ACR components at Week 14, Week 24, and Week 52; randomized patients

	Placebo + MTX (N=197) ^a	IV SIMPONI IV + MTX (N=395) ^a	p-value
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^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

^b These results are not reported as all patients received SIMPONI IV + MTX after Week 24 due to study design.

Number of swollen joints: the number of swollen joints (0-66) were counted.

Number of tender joints: the number of tender joints (0-68) were counted.

Patient's assessment of pain: patients were asked to assess their average pain during the previous week on a VAS. The scale ranged from 0 (no pain) to 10 (worst possible pain) cm.

Patient's global assessment of disease activity: patients evaluated disease activity on a VAS global assessment of disease activity. The scale ranged from 0 (very well) to 10 (very poor) cm.

Physician's global assessment of disease activity: physicians evaluated disease activity on a VAS global assessment of disease activity. The scale ranged from 0 (no arthritis activity) to 10 (extremely active arthritis) cm.

HAQ: Disability Index of the Health Assessment Questionnaire assesses the degree of difficulty in 8 functional areas (dressing, arising, eating, walking, hygiene, reaching, gripping, and activities of daily living). Improvement in HAQ scores (range 0-3) were calculated so that the positive values indicate improvement (i.e., less disability) and negative values indicate worsening.

CRP: (Normal Range 0.0-0.60 mg/dL).

Radiographic response

In GO-BEFORE the change from baseline in the vdH-S score, a composite score of structural damage that radiographically measures the number and size of joint erosions and the degree of joint space narrowing in hands/wrists and feet, was used to assess the degree of structural damage. Key results for the SIMPONI 50 mg dose at week 52 are presented in Table 6.

The number of patients with no new erosions or a change from baseline in total vdH-S Score ≤ 0 was significantly higher in the SIMPONI treatment group than in the control group ($p = 0.003$). The radiographic effects observed at week 52 were maintained through week 104. Among patients remaining in the study and treated with SIMPONI, radiographic effects were similar from week 104 through week 256.

Table 6: Radiographic mean (SD) changes from baseline in total vdH-S score at week 52 in the overall population of GO-BEFORE

	Placebo + MTX	SIMPONI 50 mg + MTX
n ^a	160	159
Total Score		
Baseline	19.7 (35.4)	18.7 (32.4)
Change from baseline	1.4 (4.6)	0.7 (5.2)*
Erosion Score		
Baseline	11.3 (18.6)	10.8 (17.4)
Change from baseline	0.7 (2.8)	0.5 (2.1)
JSN Score		
Baseline	8.4 (17.8)	7.9 (16.1)
Change from baseline	0.6 (2.3)	0.2 (2.0)**

^a n reflects randomized patients

* p = 0.015

** p = 0.044

In IV RA Study 1, structural joint damage was assessed radiographically and expressed as a change in van der Heijde-Modified Sharp Score (vdH-S), at Week 24 compared to baseline. The SIMPONI IV + MTX treatment group significantly inhibited the progression of structural damage compared with placebo + MTX, as assessed by total vdH-S score as shown in Table 7. Inhibition of radiographic progression continued to be observed in patients receiving SIMPONI IV + MTX at Week 52.

Table 7: IV RA Study 1: Radiographic change from baseline at Week 24 and Week 52

	Placebo + MTX (N=197)^a	SIMPONI IV + MTX (N=395)^a
	Mean (SD)	Mean (SD)
Total vdH-S score		
Baseline	50.3 (± 59.8)	47.6 (± 54.6)
Change from baseline at Week 24	1.1 (± 3.2)	0.03 (± 1.9)*
Change from baseline at Week 52	1.2 (± 4.0) ^b	0.13 (± 3.1)
Erosion score		
Baseline	25.6 (± 32.3)	23.9 (± 29.0)
Change from baseline at Week 24	0.5 (± 2.1)	-0.1 (± 1.1)*
Change from baseline at Week 52	0.4 (± 2.4) ^b	-0.2 (± 1.8)
JSN score		
Baseline	24.6 (± 29.5)	23.7 (± 28.3)
Change from baseline at Week 24	0.6 (± 1.7)	0.1 (± 1.3)*
Change from baseline at Week 52	0.8 (± 2.2) ^b	0.3 (± 2.0)

^a N reflects randomized patients.

^b These results are based on patients who received SIMPONI IV + MTX for at least 6 months.

* p ≤ 0.002.

At Week 24, a significantly greater proportion of patients in the SIMPONI IV + MTX group (71%) had no progression of structural damage (change in the total vdH-S score ≤ 0), compared to 57% of patients in the placebo + MTX group (Table 8).

Table 8: IV RA Study 1: Number of patients with change from baseline in total vdH-S score ≤ 0 (Week 24 and Week 52)

	Placebo + MTX (N=197)	SIMPONI IV + MTX (N=395)	p-value
Patients with change from baseline in total vdH-S score ≤ 0 at Week 24	113 (57%)	279 (71%)	0.001
Patients with change from	117 (59%) ^a	278 (70%)	

Table 8: IV RA Study 1: Number of patients with change from baseline in total vdH-S score ≤ 0 (Week 24 and Week 52)

	Placebo + MTX (N=197)	SIMPONI IV + MTX (N=395)	p-value
baseline in total vdH-S score ≤ 0 at Week 52			

^a These results are based on patients who received SIMPONI IV + MTX for at least 6 months.

Values are number (%).

Physical function and health-related quality of life

Physical function and disability were assessed as a separate endpoint in GO-FORWARD and GO-AFTER using the disability index of the HAQ DI. In these studies, SIMPONI demonstrated clinically meaningful and statistically significant improvement in HAQ DI from baseline versus control at week 24. Among patients who remained on the SIMPONI treatment to which they were randomised at study start, improvement in HAQ DI was maintained through week 104. Among patients remaining in the study and treated with SIMPONI, improvement in HAQ DI was similar from week 104 through week 256.

In GO-FORWARD clinically meaningful and statistically significant improvements were demonstrated in health-related quality of life as measured by the physical component score of the SF-36 in patients treated with SIMPONI versus placebo at week 24. Among patients who remained on the SIMPONI treatment to which they were randomised at study start, improvement of the SF-36 physical component was maintained through week 104. Among patients remaining in the study and treated with SIMPONI, improvement of the SF-36 physical component was similar from week 104 through week 256. In GO-FORWARD and GO-AFTER, statistically significant improvements were observed in fatigue as measured by functional assessment of chronic illness therapy-fatigue scale (FACIT-F).

In IV RA Study 1, physical function and health-related quality of life were assessed using the disability index of the HAQ, the SF-36 health survey, and the Functional Assessment of Chronic Illness Therapy - Fatigue scale (FACIT-F). Patients treated with SIMPONI IV + MTX showed significantly greater median improvements in the HAQ compared with placebo IV + MTX at Week 14 (median 0.50 vs. 0.12; $p < 0.001$) and Week 24 (median 0.50 vs. 0.12; $p < 0.001$). The mean ($\pm$ SD) improvement in the HAQ from baseline to Week 14 was 0.50 ± 0.58 for the SIMPONI IV + MTX group and 0.19 ± 0.56 for the placebo IV + MTX group. The mean improvement from baseline to Week 24 was 0.53 ± 0.64 for the SIMPONI IV + MTX group and 0.20 ± 0.55 for the placebo group (see Table 9). The mean improvement in HAQ was maintained through Week 52. The proportion of patients achieving a clinically meaningful ≥ 0.25 improvement in HAQ from baseline at Week 14 was greater in the SIMPONI IV + MTX group than in the placebo IV + MTX group (68% compared with 43%; $p < 0.001$). This response was maintained at Week 24 (67% in the SIMPONI IV + MTX group vs. 45% in the placebo IV + MTX group; $p < 0.001$) and through Week 52.

Table 9: IV RA Study 1: Improvement in physical function as measured by HAQ

	Placebo IV + MTX (N=197)^a	SIMPONI IV + MTX* (N=395)^a
HAQ baseline score		
Mean $\pm$ SD	1.57 $\pm$ 0.62	1.56 $\pm$ 0.67
Median	1.63	1.63

Table 9: IV RA Study 1: Improvement in physical function as measured by HAQ

	Placebo IV + MTX (N=197)^a	SIMPONI IV + MTX* (N=395)^a
Improvement in HAQ		
Week 14		
Mean $\pm$ SD	0.19 $\pm$ 0.56	0.50 $\pm$ 0.58
Median	0.12	0.50
Week 24		
Mean $\pm$ SD	0.20 $\pm$ 0.55	0.53 $\pm$ 0.64
Median	0.12	0.50
Week 52		
Mean $\pm$ SD	N/A ^b	0.51 $\pm$ 0.65
Median	N/A ^b	0.38

* p<0.001 for all comparisons; p-value calculations are based on comparisons of median values.

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

Improvement in HAQ scores (range 0-3) were calculated so that the positive values indicate improvement (i.e., less disability) and negative values indicate worsening.

^b These results are not reported as all patients received SIMPONI IV + MTX after Week 24 due to study design.

In IV RA Study 1, median changes from baseline in both the SF-36 physical component summary scores and mental component summary scores were significantly greater with SIMPONI IV + MTX compared with patients treated with placebo IV + MTX at Week 12, Week 16, and Week 24 (p<0.001 for each component at all time points). Improvements in SF-36 physical component scores and mental component scores were maintained through Week 52.

SIMPONI IV + MTX showed significantly greater median improvement from baseline in the SF-36 physical component summary (PCS) score compared with placebo IV + MTX at Week 12 (5.6 vs. 2.6; p<0.001), Week 16 (7.1 vs. 1.9; p<0.001) and Week 24 (7.4 vs. 3.3; p<0.001). At baseline, the mean SF-36 PCS scores of 30.8 ± 6.8 in the SIMPONI IV + MTX group and 30.9 ± 7.3 in the placebo IV + MTX group were lower than the norm of 50 ± 10 measured in the US population. The mean improvement from baseline in the PCS score of the SF-36 at Week 12 was greater in the SIMPONI IV + MTX group than in the placebo IV + MTX group (5.9 ± 7.7 vs. 3.2 ± 7.4 , respectively). The mean improvement on the SF-36 PCS score was maintained through Week 16, Week 24, and Week 52.

SIMPONI IV + MTX showed significantly greater median improvement from baseline in the SF-36 mental component summary (MCS) score compared with placebo IV + MTX at Week 12 (4.3 vs. 0.6; p<0.001), Week 16 (6.4 vs. 1.9; p<0.001) and Week 24 (6.4 vs. 1.1; p<0.001). The mean change from baseline in the MCS score of the SF-36 at Week 12 was greater in the SIMPONI IV + MTX group than in the placebo IV + MTX group (4.9 ± 10.3 vs. 1.5 ± 9.9 , respectively). The mean improvement on the SF-36 MCS score was maintained through Week 16, Week 24, and Week 52.

In addition, the eight scales of the SF-36 that comprise the PCS and MCS scores showed greater median improvements at Week 12, Week 16, and Week 24 for the SIMPONI IV + MTX group compared to the placebo IV + MTX group (p<0.001 for all weeks and all scales). Compared to Week 24, improvements in each of the eight SF-36 scales were all maintained at Week 52.

Patients treated with SIMPONI IV + MTX had a clinically significant median improvement in fatigue as measured by FACIT-F. The median change from baseline in the FACIT-F score was significantly greater for SIMPONI IV + MTX compared with the placebo IV + MTX group at Week 16 (7.0 vs. 2.0; $p < 0.001$) and Week 24 (8.0 vs. 2.0; $p < 0.001$). Improvements in FACIT-F observed at Week 24 were maintained at Week 52. The median change from baseline in FACIT-F through Week 16 and Week 24 in the SIMPONI IV + MTX group was ≥ 4 points, which indicated a clinically meaningful improvement on fatigue. The mean change from baseline in the FACIT-F scores was greater in the SIMPONI IV + MTX group than in the placebo IV + MTX group at Week 16 (7.54 ± 10.546 vs. 2.16 ± 9.700 , respectively). The mean improvement in the FACIT-F score was maintained through Week 24 in the SIMPONI IV + MTX group when compared with placebo (7.96 ± 10.793 vs. 2.54 ± 10.219 respectively, $p < 0.001$). In addition, the proportion of patients achieving a clinically meaningful improvement (≥ 4 points) for FACIT-F at Week 24 was maintained at Week 52.

Health economics

Health economics data on health care resource utilization, time lost from work by patients and caregivers, employability, and patient productivity at work, school, or home were collected through questionnaires at baseline and every 8 weeks thereafter. Patients were asked to indicate how much their disease affected their productivity in the previous 4 weeks using a 0 to 10 cm VAS scale (not at all affected to affected very much). Only those assessments where statistically significant differences were observed between SIMPONI $\pm$ MTX and placebo $\pm$ MTX are presented below.

IV RA study 1

The median change from baseline in health state VAS scale (EQ VAS) was significantly greater in the SIMPONI IV + MTX group than in the placebo IV + MTX group at Week 16 and Week 24. At Week 24, the median improvement from baseline in EQ VAS scale was 20.00 in the SIMPONI + MTX group compared with 6.00 in the placebo + MTX group ($p < 0.001$), and exceeded the threshold of clinically meaningful change defined as $\frac{1}{2}$ standard deviation of baseline value ($24.86/2=12.5$).

Median decrease in self-reported impact of disease on productivity was significantly greater in the SIMPONI IV + MTX group compared with the placebo IV + MTX group at Week 24 (-2.9 vs. -0.5 ; $p < 0.001$). The mean $\pm$ SD decrease in self-reported impact of disease on productivity in the SIMPONI IV + MTX group and placebo IV + MTX group at Week 24 was -2.78 ± 2.92 vs. -1.03 ± 2.96 , respectively ($p < 0.001$).

Psoriatic arthritis

The safety and efficacy of SIMPONI were evaluated in a multi-centre, randomized, double-blind, placebo-controlled study (GO-REVEAL) in 405 adult patients with active PsA (≥ 3 swollen joints and ≥ 3 tender joints) despite non-steroidal anti-inflammatory (NSAID) or DMARD therapy. Patients in this study had a diagnosis of PsA for at least 6 months and had at least mild psoriatic disease. Patients with each sub-type of psoriatic arthritis were enrolled, including polyarticular arthritis with no rheumatoid nodules (43%), asymmetric peripheral arthritis (30%), distal interphalangeal (DIP) joint arthritis (15%), spondylitis with peripheral arthritis (11%), and arthritis mutilans (1%). Previous treatment with an anti-TNF agent was not allowed. SIMPONI or placebo were administered subcutaneously every 4 weeks. Patients were randomly assigned to placebo, SIMPONI 50 mg, or SIMPONI 100 mg. Patients receiving placebo were switched to SIMPONI 50 mg after week 24. Patients entered an open label long-term extension at week 52. Approximately forty-eight percent of patients continued on stable doses of methotrexate (≤ 25 mg/week). The co-primary endpoints were the percentage of patients achieving ACR 20 response at week 14 and change from baseline in total PsA modified vdH-S score at week 24.

In general, no clinically meaningful differences in measures of efficacy were observed between the

SIMPONI 50 mg and 100 mg dosing regimens through week 104. By study design, patients in the long-term extension may have switched between the 50 mg and 100 mg SIMPONI doses at the discretion of the study physician.

IV PsA study

The efficacy and safety of SIMPONI IV were evaluated in a multicenter, randomized, double-blind, placebo-controlled trial (GO-VIBRANT) in 480 adults with active psoriatic arthritis despite nonsteroidal anti-inflammatory drug (NSAID) or disease-modifying antirheumatic drug (DMARD) therapy. Patients in this trial had a diagnosis of PsA for at least six months and had symptoms of active disease (≥ 5 swollen joints, ≥ 5 tender joints, and a CRP level of ≥ 0.6 mg/dL). Patients were randomized to receive SIMPONI IV 2 mg/kg (N=241) or placebo (N=239) as a 30-minute intravenous infusion at Weeks 0, 4, 12 and 20. All patients on placebo received SIMPONI IV at Week 24, Week 28 and every 8 weeks thereafter through Week 52. Patients in the group treated with SIMPONI IV continued to receive infusions of SIMPONI IV at Week 28 and every 8 weeks through Week 52. Previous treatment with a biologic was not allowed.

Patients were allowed to continue stable doses of MTX, NSAIDs, and low dose oral corticosteroids (equivalent to ≤ 10 mg of prednisone per day) during the study. At study enrollment, the use of other DMARDs including cytotoxic agents or other biologics was prohibited.

Patients with each subtype of PsA were enrolled, including polyarticular arthritis with absence of rheumatoid nodules (44%), asymmetric peripheral arthritis (19%), distal interphalangeal joint involvement (8.1%), spondylitis with peripheral arthritis (25%), and arthritis mutilans (4.8%). The median duration of PsA disease was 3.5 years, 86% of patients had previously used MTX, and 35% of patients received at least one other DMARD in the past. At baseline, 76% and 54% of the patients had enthesitis and dactylitis, respectively. During the trial, the use of concomitant medications was MTX (70%), oral corticosteroids (28%), and NSAIDs (71%).

The primary endpoint was the percentage of patients achieving an ACR 20 response at Week 14. The major secondary endpoints were change in baseline in the HAQ-DI score at Week 14, the proportion of subjects who achieve an ACR 50 response at Week 14, the proportion of subjects (with baseline $\geq 3\%$ Body Surface Area [BSA] psoriatic involvement) who achieve a PASI 75 response at Week 14, and the change from baseline in total modified vdH-S score at Week 24.

Signs and symptoms

Key results for the 50 mg dose at weeks 14 and 24 are shown in Table 10 and described below.

Table 10: Key efficacy outcomes from GO-REVEAL

	Placebo	SIMPONI 50 mg*
n ^a	113	146
Responders, % of patients		
ACR 20		
Week 14	9 %	51 %
Week 24	12 %	52 %
ACR 50		
Week 14	2 %	30 %
Week 24	4 %	32 %
ACR 70		
Week 14	1 %	12 %

Week 24	1 %	19 %
PASI^b 75^c		
Week 14	3 %	40 %
Week 24	1 %	56 %
* p < 0.05 for all comparisons; ^a n reflects randomised patients; actual number of patients evaluable for each endpoint may vary by timepoint ^b <i>Psoriasis Area and Severity Index</i> ^c Based on the subset of patients with ≥ 3% BSA involvement at baseline, 79 patients (69.9%) in the placebo group and 109 (74.3%) in the SIMPONI 50 mg group.		

Responses were observed at the first assessment (week 4) after the initial SIMPONI administration. Similar ACR 20 responses at week 14 were observed in patients with polyarticular arthritis with no rheumatoid nodules and asymmetric peripheral arthritis PsA subtypes. The number of patients with other PsA subtypes was too small to allow meaningful assessment. Responses observed in the SIMPONI treated groups were similar in patients receiving and not receiving concomitant MTX. Among 146 patients randomised to SIMPONI 50 mg, 70 were still on this treatment at week 104. Of these 70 patients, 64, 46 and 31 patients had an ACR 20/50/70 response, respectively. Among patients remaining in the study and treated with SIMPONI, similar rates of ACR 20/50/70 response was observed from week 104 through week 256.

Statistically significant responses in DAS28 were also observed at weeks 14 and 24 (p < 0.05).

At week 24, improvements in parameters of peripheral activity characteristic of psoriatic arthritis (e.g. number of swollen joints, number of painful/tender joints, dactylitis and enthesitis) were seen in the SIMPONI-treated patients. SIMPONI treatment resulted in significant improvement in physical function as assessed by HAQ DI, as well as significant improvements in health-related quality of life as measured by the physical and mental component summary scores of the SF-36. Among patients who remained on the SIMPONI treatment to which they were randomised at study start, DAS28 and HAQ DI responses were maintained through week 104. Among patients remaining in the study and treated with SIMPONI, DAS28 and HAQ DI responses were similar from week 104 through week 256.

IV PsA study

Treatment with SIMPONI IV, compared with placebo, resulted in a significant improvement in signs and symptoms as demonstrated by the percentage of patients with an ACR 20 response at Week 14 (see Table 11). In the SIMPONI IV-treated group, 75% of patients achieved an ACR 20 response compared with 22% in the placebo group (p<0.001). This benefit was consistently observed when assessed in SIMPONI IV-treated patients across the five PsA subtypes. ACR 20 responses observed in the SIMPONI IV-treated groups were similar in patients who were receiving and not receiving concomitant MTX. In addition, the percentage of patients achieving an ACR 20 response at Week 24 was significantly greater for patients receiving SIMPONI IV (77%) compared with patients receiving placebo (24%) (p<0.001) (Table 11).

The percentage of patients achieving ACR 50, ACR 70 and ACR 90 was greater in patients treated with SIMPONI IV compared with placebo (Table 11). The percentage of patients achieving an ACR 50 response at Week 14 was 44% for patients treated with SIMPONI IV and 6.3% for patients treated with placebo (p<0.001). At Week 24, the percentage of patients achieving an ACR 50 response was 54% and 6.3% (p<0.001) for SIMPONI IV and placebo groups, respectively. The percentage of patients achieving an ACR 70 response at Week 14 was 25% for patients treated with SIMPONI IV and 2.1% for patients treated with placebo (p<0.001). At Week 24, the percentage of patients achieving an ACR 70 response was

33% and 3.3% ($p<0.001$) for SIMPONI IV and placebo groups, respectively. The percentage of patients achieving an ACR 90 response at Week 14 was 3.7% for patients treated with SIMPONI IV and 0.4% for patients treated with placebo ($p=0.011$). At Week 24, the percentage of patients achieving an ACR 90 response was 6.2% and 0% ($p<0.001$) for SIMPONI IV and placebo groups, respectively (Table 11).

Table 11: IV PsA study: Percentage of Patients with ACR responses at Weeks 14 and 24: randomized patients

	Placebo N=239^a	SIMPONI IV N=241^a	Treatment Difference (95% CI)
ACR 20 (% responders)			
Week 14	22%	75%	53%* (45.8, 60.9)
Week 24	24%	77%	53%* (44.9, 60.1)
ACR 50 (% responders)			
Week 14	6.3%	44%	37%* (30.3, 44.2)
Week 24	6.3%	54%	47%* (40.2, 54.2)
ACR 70 (% responders)			
Week 14	2.1%	25%	22%* (16.7, 28.1)
Week 24	3.3%	33%	29%* (23.1, 35.8)
ACR 90 (% responders)			
Week 14	0.4%	3.7%	3.3%** (0.80, 5.8)
Week 24	0	6.2%	6.2%* (3.2, 9.3)

^a N reflects randomized patients.

Bold text indicates primary endpoint.

* $p<0.001$

** $p=0.011$

The percentage of patients achieving ACR 20 responses by visit through Week 24 for the IV PsA study is shown in Figure 2. After the initial administration, ACR 20 responses were observed in 46% of SIMPONI IV-treated patients at the first assessment (Week 2) compared with 7.5% of placebo-treated patients. Responses for ACR 50 and ACR 70 were also observed as early as the first assessment (Week 2).

Figure 2: IV PsA Study: Percentage of Patients Achieving ACR 20 Response Through Week 24: randomized patients

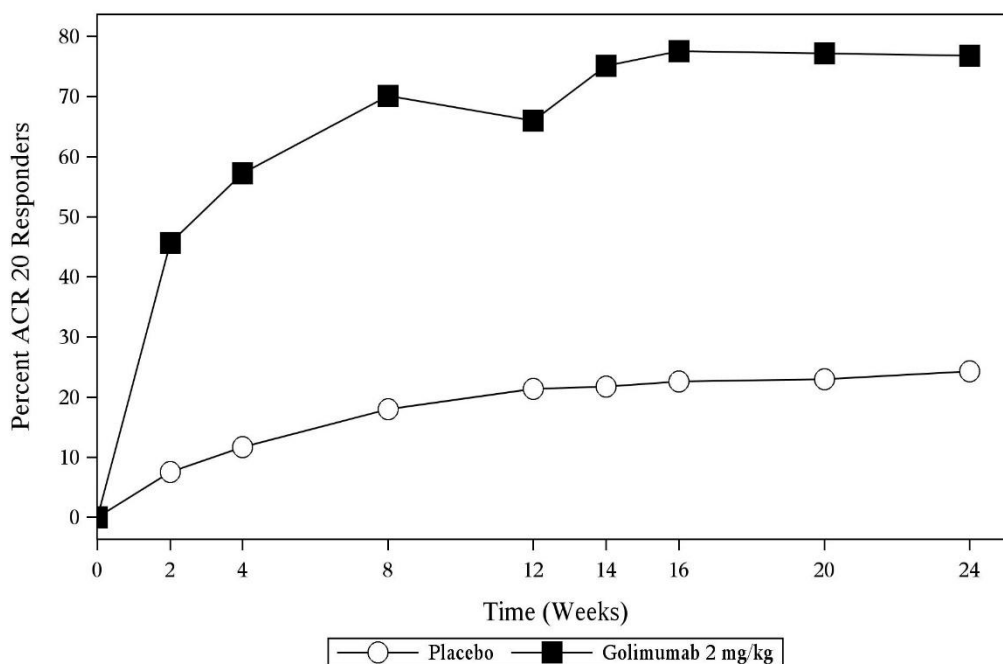


Table 12 shows the median percent improvement in the components of the ACR response criteria for the SIMPONI IV and placebo groups in the IV PsA study.

Table 12: IV PsA Study: Median Percent improvement from baseline in ACR components at Week 14 and Week 24, randomized patients

	Placebo N=239^a	SIMPONI IV N=241^a
ACR Components		
Number of swollen joints (0-66)		
Baseline	12	11
Week 14	32%	87%
Week 24	43%	90%
Number of tender joints (0-68)		
Baseline	24	22
Week 14	15%	71%
Week 24	23%	79%
Patient's assessment of pain (0-100 mm)		
Baseline	67	67
Week 14	12%	53%
Week 24	15%	62%
Patient global assessment of disease activity (0-100 mm)		
Baseline	66	68
Week 14	13%	51%
Week 24	12%	59%
Physician global assessment disease activity (0-100 mm)		
Baseline	65	64
Week 14	16%	70%
Week 24	24%	75%
Disability Index (HAQ) (0-3)^b		
Baseline	1.2	1.2
Week 14	10%	44%
Week 24	11%	50%
CRP (mg/L)^c		
Baseline	14	12
Week 14	22%	86%
Week 24	21%	83%

Note: All values are medians.

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary.

^b Health Assessment Questionnaire-Disability Index.

^c CRP = C-reactive protein

Table 13 shows the median change from baseline at Week 14 in the components of the ACR response criteria for the SIMPONI IV and placebo groups in the IV PsA study.

Table 13: IV PsA Study: Median Change from Baseline of ACR Components at Week 14: randomized subjects

	Placebo N=239 ^a		SIMPONI IV N=241 ^a	
	Baseline	Week 14 change from baseline	Baseline	Week 14 change from baseline
ACR Components				
Number of swollen joints (0-66)	12	-4.0	11	-8.0
Number of tender joints (0-68)	24	-3.0	22	-13
Patient's assessment of pain (0-100 mm)	67	-8.0	67	-33
Patient global assessment of disease activity (0-100 mm)	66	-9.0	68	-33
Physician global assessment of disease activity (0-100 mm)	65	-11	64	-39
Disability Index (HAQ) (0-3) ^b	1.2	-0.13	1.2	-0.63*
CRP (mg/L) ^c	14	-2.1	12	-9.2

Note: All values are medians.

* p<0.001.

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary.

^b Health Assessment Questionnaire-Disability Index.

^c CRP = C-reactive protein

The percentage of patients achieving modified Psoriatic Arthritis Response Criteria (mPsARC), Disease Activity Score including 28 joints using C-reactive protein (DAS28) response, and DAS28 remission was also greater in patients treated with SIMPONI IV compared with placebo at Week 14 and Week 24 (Table 14). The percentage of patients achieving an mPsARC response at Week 14 and Week 24 was 84% for patients treated with SIMPONI IV and 34% for patients treated with placebo (p<0.001). At baseline, the mean DAS28 score in the SIMPONI IV-treated group and the placebo-treated group was 5.4 and 5.5, respectively. The percentage of patients achieving a DAS28 response at Week 14 was 93% for patients treated with SIMPONI IV and 40% for patients treated with placebo (p<0.001). At Week 24, the percentage of patients achieving a DAS28 response was 93% and 40% (p<0.001) for SIMPONI IV and placebo groups, respectively. The percentage of patients achieving DAS28 remission at Week 14 was 37% for patients treated with SIMPONI IV and 4.6% for patients treated with placebo (p<0.001). At Week 24, the percentage of patients achieving DAS28 remission was 42% and 7.1% (p<0.001) for SIMPONI IV and placebo groups, respectively. The mean change from baseline in DAS28 score at Week 14 was -2.3 and -0.6 in patients treated with SIMPONI IV and placebo, respectively. At Week 24, the mean change from baseline in DAS28 score was -2.50 and -0.7 in patients treated with SIMPONI IV and placebo, respectively. A negative/decreasing DAS28 score is indicative of improvement. Responses for mPsARC and DAS28 were observed as early as the first assessment (Week 2).

Table 14: IV PsA study: Percentage of Patients with modified PsARC Response, DAS28 Response, and DAS28 Remission at Weeks 14 and 24: randomized patients

	Placebo N=239 ^a	SIMPONI IV N=241 ^a	Treatment Difference (95% CI)
Modified PsARC^b (% responders)			
Week 14	34%	84%	50% (42.3, 57.6)
Week 24	34%	84%	50% (41.9, 57.1)
DAS28 (CRP) Response (% responders)			
Week 14	40%	93%	52% (45.3, 59.4)
Week 24	40%	93%	53% (46.2, 60.2)
DAS28 (CRP) Remission (% remitters)			
Week 14	4.6%	37%	32% (25.3, 38.6)
Week 24	7.1%	42%	35% (27.8, 41.8)

^a N reflects randomized patients.

^b Modified Psoriatic Arthritis Responder Criteria (PsARC) is defined as improvement in at least 2 of the following criteria:

- ≥ 30% decrease in the swollen joint count
- ≥ 30% decrease in the tender joint count
- ≥ 20% improvement in the patient's overall assessment on a visual analog scale; and
- ≥ 20% improvement in the physician's overall assessment on a visual analog scale
- including at least one of the joint criteria and with no deterioration in the other criteria.

Table 15 shows the improvement in the enthesitis and dactylitis for the SIMPONI IV and placebo groups in the IV PsA study. Patients with enthesitis at baseline were evaluated for improvement using the Leeds Enthesitis Index (LEI) on a scale of 0-6. At baseline, the median LEI score in patients with enthesitis in the SIMPONI IV-treated group and the placebo-treated group was 3.0 and 3.0, respectively. SIMPONI IV-treated patients showed a median improvement in LEI score of 2.0 as compared with a median improvement in placebo-treated patients of 0.0 at Week 14 ($p < 0.001$). Patients with dactylitis at baseline were evaluated for improvement on a scale of 0-60. At baseline, the median dactylitis score in patients with dactylitis in the SIMPONI IV-treated group and the placebo-treated group was 5.5 and 6.0, respectively. SIMPONI IV-treated patients showed a median improvement in dactylitis score of 4.0 compared with a median improvement of 2.0 in placebo-treated patients at Week 14 ($p < 0.001$). Improvements in enthesitis and dactylitis were maintained in SIMPONI IV-treated group through Week 24.

Table 15: IV PsA Study: Median improvement from baseline and percent improvement in enthesitis score in patients with enthesitis at baseline and median improvement from baseline and percent improvement in dactylitis score in patients with dactylitis at baseline: randomized patients

	Placebo N=239 ^a	SIMPONI IV N=241 ^a
Enthesitis score (median)		
Patients with enthesitis at baseline	181	185
Baseline score	3.0	3.0
Improvement from Baseline		

Week 14	0.0	2.0*
Week 24	1.0	2.0
Percent Improvement		
Week 14	0%	83%
Week 24	33%	100%
Dactylitis score (median)		
Patients with dactylitis at baseline	124	134
Baseline	6.0	5.5
Improvement from Baseline		
Week 14	2.0	4.0*
Week 24	2.0	4.0
Percent Improvement		
Week 14	21%	100%
Week 24	54%	100%

Note: All values are medians.

* p<0.001

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary.

Enthesitis: Leeds Enthesitis Index (LEI) assesses enthesitis in patients with PsA, and evaluates the presence (score of 1) or absence (score of 0) of pain by applying pressure at the following entheses: lateral elbow epicondyle (left and right), medial femoral condyle (left and right), and Achilles tendon insertion (left and right). Patients are scored on a scale of 0-6.

Dactylitis: Dactylitis was assessed in 20 digits (10 fingers and 10 toes) using a scoring system from 0 to 3 (no dactylitis to severe dactylitis). The dactylitis score was calculated as the sum of scores of all 20 digits (range 0 to 60). A patient is considered to have digits with dactylitis if the score is > 0.

Patients are classified as achieving psoriatic arthritis minimal disease activity (MDA) if five of the following seven outcome measures are fulfilled: tender joint count ≤ 1 ; swollen joint count ≤ 1 ; PASI ≤ 1 or BSA ≤ 3 ; patient pain visual analog scale (VAS) score of ≤ 15 ; patient global disease activity VAS score of ≤ 20 ; HAQ score ≤ 0.5 ; and tender entheses points ≤ 1 . Greater proportions of patients in the SIMPONI IV-treated group compared with placebo achieved minimal disease activity responses at Weeks 14 (27% vs 4.2%) and 24 (34% vs 4.6%).

Radiographic response

Structural damage in both hands and feet was assessed radiographically by the change from baseline in the vdH-S score, modified for PsA by addition of hand distal interphalangeal (DIP) joints.

SIMPONI 50 mg treatment reduced the rate of progression of peripheral joint damage compared with placebo treatment at week 24 as measured by change from baseline in total modified vdH-S Score (mean $\pm$ SD score was 0.27 ± 1.3 in the placebo group compared with -0.16 ± 1.3 in the SIMPONI group; $p = 0.011$). Out of 146 patients who were randomized to SIMPONI 50 mg, 52 week X-ray data were available for 126 patients, of whom 77% showed no progression compared to baseline. At week 104, X-ray data were available for 114 patients, and 77% showed no progression from baseline. Among patients remaining in the study and treated with SIMPONI, similar rates of patients showed no progression from baseline from week 104 through week 256.

IV PsA study

Structural joint damage was assessed radiographically and expressed as a change in total modified van der Heijde-Sharp (vdH-S) score and its components, the erosion score and joint space narrowing (JSN) score, at Week 24 compared to baseline. At baseline, the mean total modified vdH-S score was 35.0, the mean

erosion score was 21.7, and the mean JSN score was 13.3. No change from baseline or a decrease from baseline in total modified vdH-S, erosion, or JSN scores is indicative of the inhibition of progression. At Week 24, SIMPONI IV significantly inhibited the progression of structural damage compared with placebo, as assessed by total modified vdH-S score and its components as shown in Table 16.

Table 16: IV PsA Study 1: Radiographic Change from Baseline at Week 24

	Placebo N=237		SIMPONI IV N=237	
	Baseline	Week 24 Change from baseline	Baseline	Week 24 Change from baseline
Total Modified vdH-S Score	34.4 (53.5)	2.0 (0.3)	35.5 (55.1)	-0.4 (0.1)*
Erosion Score	21.1 (30.2)	1.3 (2.8)	22.2 (31.7)	-0.3 (1.7)*
JSN Score	13.3 (24.3)	0.6 (1.7)	13.2 (24.8)	-0.1 (1.1)*

Note: All values are means (standard deviation).

* p<0.001

Among patients with ≥ 1 joint with a JSN score of 0 at baseline (92%), a greater proportion of SIMPONI IV-treated patients showed no new erosions and no new joint space narrowing compared with placebo-treated patients at Week 24 (see Table 17).

Table 17: IV PsA Study: Percentage of Patients with New erosions and JSN in previously uninvolved joints (Week 24)

	Placebo N=237	SIMPONI IV N=237	p-value
Patients with at least 1 previously uninvolved joint	213	230	
Patients with no new erosions	115 (54%)	171 (74%)	<0.001
Patients with at least 1 previously uninvolved joint	214	229	
Patients with no new JSN	162 (76%)	205 (90%)	<0.001

Values are number (%).

At Week 24, a greater proportion of patients in the SIMPONI IV-treated group (72%) had no progression of structural damage (change in the total modified vdH-S score ≤ 0), compared to 43% of patients in the placebo group (p<0.001) (Table 18).

Table 18: IV PsA Study: Number of patients with change from baseline in total PsA modified vdH S score ≤ 0 (Week 24)

	Placebo N=237	SIMPONI IV N=237	p-value
Patients with change from baseline in total PsA modified vdH-S score ≤ 0	102 (43%)	170 (72%)	<0.001

Values are number (%).

Improvement in physical function and health-related quality of life

IV PsA Study

Physical function and health-related quality of life were assessed using the disability index of the HAQ (HAQ-DI), the SF-36 health survey, EQ 5D-5L (EuroQoL 5 Dimensions questionnaire) Index Score, EQ VAS Score, and the Functional Assessment of Chronic Illness Therapy-Fatigue (FACIT-Fatigue) questionnaire.

Patients treated with SIMPONI IV showed significantly greater median improvement from baseline in the HAQ-DI score compared with placebo at Week 14 (0.63 vs. 0.13; $p < 0.001$) and Week 24 (0.63 vs. 0.13; $p < 0.001$). Improvement in physical function as assessed by HAQ-DI demonstrated that the proportion of patients who achieved clinically meaningful improvement of ≥ 0.3 in HAQ-DI score from baseline was greater in the SIMPONI IV-treated group compared to placebo at Week 14 (69% vs. 32%; $p < 0.001$) and Week 24 (69% vs. 33%; $p < 0.001$) (see Table 19).

Table 19: IV PsA Study: Improvement in physical function as measured by HAQ; randomized patients

	Placebo N=236^a	SIMPONI IV N=237^a
HAQ baseline score^b		
Mean $\pm$ SD	1.26 $\pm$ 0.58	1.31 $\pm$ 0.56
Median	1.25	1.25
Improvement from baseline		
Week 14		
Mean $\pm$ SD	0.12 $\pm$ 0.47	0.60 $\pm$ 0.53
Median	0.13	0.63
Week 24		
Mean $\pm$ SD	0.14 $\pm$ 0.49	0.63 $\pm$ 0.54
Median	0.13	0.63
HAQ ≥ 0.3 improvement (% responders)		
Week 14	32%	69%
Week 24	33%	69%
HAQ ≥ 0.35 improvement (% responders)^b		
Week 14	32%	69%
Week 24	33%	69%

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

^b Health Assessment Questionnaire-Disability Index assesses the degree of difficulty in 8 functional areas (dressing, arising, eating, walking, hygiene, reaching, gripping, and activities of daily living) and scored from 0 (no difficulty) to 3 (inability to perform a task) with lower scores indicating better functioning. In PsA, an improvement in score of 0.30 has been determined to indicate a clinically meaningful improvement.

Patients treated with SIMPONI IV showed significantly greater median improvement from baseline in the SF-36 physical component summary (PCS) score compared with placebo at Week 14 (8.7 vs. 1.8; $p < 0.001$) and Week 24 (9.3 vs. 1.5). At baseline, the median SF-36 PCS scores were 33 in the SIMPONI IV group and 34 for the placebo group. Greater proportions of subjects in the SIMPONI IV group compared with placebo achieved a clinically meaningful change from baseline in SF-36 PCS score (an increase of 5 or more units) from Week 8 (63% vs 26%) through Week 24 (70% vs 29%).

Patients treated with SIMPONI IV showed significantly greater median improvement from baseline in the

SF-36 mental component summary (MCS) score compared with placebo at Week 14 (5.2 vs. 0.4; $p < 0.001$) and Week 24 (4.1 vs. 1.3). At baseline, the median SF-36 MCS scores were 43 in the SIMPONI IV group and 42 for the placebo group. Greater proportions of subjects in the SIMPONI IV group compared with placebo achieved a clinically meaningful change from baseline in SF-36 MCS score (an increase of 5 or more units) from Week 8 (46% vs 27%) through Week 24 (47% vs 29%).

In addition, the eight domains of the SF-36 that comprise the PCS and MCS scores showed greater median improvements at Week 8, Week 14, and Week 24 for the SIMPONI IV group compared to the placebo group.

EuroQoL 5D-5L is a measure of health status to provide a measure of health for clinical and economic appraisal. EQ-5D-5L consists of 2 elements: The EQ-5D-5L descriptive system and the EQ visual analogue scale (EQ VAS). The EQ-5D descriptive system comprises the following 5 dimensions: mobility, self-care, usual activities, pain/discomfort and anxiety/depression. EQ VAS records the respondent's self-rated health on a VAS, with the endpoints of "worst imaginable health" and "Best imaginable health".

Patients treated with SIMPONI IV achieved improvement from baseline in the median EQ-5D-5L Index Score and EQ-VAS Score compared to placebo-treated patients at Week 14 (0.14 vs. 0.02 and 20 mm vs. 4.0 mm, respectively). Median baseline values for EQ-5D-5L index score and EQ VAS score, which were comparable between placebo and SIMPONI IV arms for all subjects, were 0.6 and 48 mm, respectively (see Table 20).

Table 20: IV PsA Study: Health related quality of life; randomized patients

	Placebo (N=239)^a	SIMPONI (N=241)^a
SF-36 PCS		
Baseline score		
Mean $\pm$ SD	34 $\pm$ 7.2	33 $\pm$ 6.9
Median	34	33
Week 14, improvement from baseline		
Mean $\pm$ SD	2.7 $\pm$ 5.9	8.7 $\pm$ 7.6
Median	1.8	8.7
Week 24, improvement from baseline		
Mean $\pm$ SD	2.4 $\pm$ 6.1	9.4 $\pm$ 8.1
Median	1.5	9.3
SF-36 MCS		
Baseline score		
Mean $\pm$ SD	42 $\pm$ 10.2	43 $\pm$ 11.4
Median	42	43
Week 14, improvement from baseline		
Mean $\pm$ SD	1.0 $\pm$ 7.6	5.3 $\pm$ 10.0
Median	0.43	5.2
Week 24, improvement from baseline		
Mean $\pm$ SD	0.8 $\pm$ 7.4	5.3 $\pm$ 10.2
Median	1.3	4.1
EQ-5D Index Score^b		
Baseline score ^b		

Table 20: IV PsA Study: Health related quality of life; randomized patients

Mean $\pm$ SD	0.59 $\pm$ 0.15	0.58 $\pm$ 0.14
Median	0.62	0.60
Week 14, improvement from baseline		
Mean $\pm$ SD	0.03 $\pm$ 0.12	0.15 $\pm$ 0.16
Median	0.02	0.14
Week 24, improvement from baseline		
Mean $\pm$ SD	0.04 $\pm$ 0.13	0.16 $\pm$ 0.15
Median	0.03	0.14
EQ-5D VAS (mm)		
Baseline score (mm) ^b		
Mean $\pm$ SD	46.2 $\pm$ 20.3	46.9 $\pm$ 20.1
Median	47.5	48
Week 14, improvement from baseline (mm)		
Mean $\pm$ SD	5.3 $\pm$ 21.0	19 $\pm$ 24.3
Median	4.0	20
Week 24, improvement from baseline (mm)		
Mean $\pm$ SD	5.5 $\pm$ 23.1	20 $\pm$ 24.2
Median	4.0	20

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

^b EuroQoL 5D-5L is a measure of health status to provide a measure of health for clinical and economic appraisal. EQ-5D-5L consists of 2 elements: The EQ-5D-5L descriptive system and the EQ visual analogue scale (EQ VAS). The EQ-5D descriptive system comprises the following 5 dimensions: mobility, self-care, usual activities, pain/discomfort and anxiety/depression. EQ VAS records the respondent's self-rated health on a visual analogue scale (VAS, mm), with the endpoints of "worst imaginable health" and "Best imaginable health".

Functional Assessment of Chronic Illness Therapy - Fatigue (FACIT-F) questionnaire consists of 13 questions that assess a subject's level of fatigue and tiredness over the last 7 days on a scale from 0-52. At baseline, the median FACIT-Fatigue score in the SIMPONI IV-treated group and the placebo-treated group was 28. The median improvement from baseline in the FACIT-Fatigue score was significantly greater for SIMPONI IV compared with the placebo group at Week 14 (8.0 vs. 2.0; $p < 0.001$) and Week 24 (8.0 vs. 3.0; $p < 0.001$) (see Table 21). The median change from baseline in FACIT-Fatigue through Week 14 and Week 24 in the SIMPONI IV group was ≥ 4 points, which indicated a clinically meaningful improvement in fatigue. In addition, a greater proportion of SIMPONI IV-treated patients achieved a clinically meaningful improvement (≥ 4 points) from baseline for FACIT-Fatigue score compared with placebo at Weeks 14 and 24 (68% vs. 41% and 70% vs. 43%, respectively).

Table 21: IV PsA Study: Improvement in fatigue as measured by FACIT-F; randomized patients

	Placebo N=239^a	SIMPONI IV N=241^a
Baseline FACIT-F score ^b		
Mean $\pm$ SD	27.7 $\pm$ 9.7	27.9 $\pm$ 9.6
Median	28.0	28.0
Week 14, improvement from baseline		
Mean $\pm$ SD	2.2 $\pm$ 7.6	8.4 $\pm$ 9.9
Median	2.0	8.0
Week 24, improvement from baseline		
Mean $\pm$ SD	2.3 $\pm$ 7.8	9.2 $\pm$ 9.8
Median	3.0	8.0

^a N reflects randomized patients; actual number of patients evaluable for this endpoint may vary by timepoint.

^b Functional Assessment of Chronic Illness Therapy - Fatigue (FACIT-F) questionnaire consists of 13 questions that assess a subject's level of fatigue and tiredness over the last 7 days as measured by mean change from baseline score on a scale from 0-52

Health economics

IV PsA Study

Health economics data on patient productivity at work was collected through the Work Limitations Questionnaire (WLQ) and productivity visual analog scale (VAS) at baseline and Weeks 8, 14 and 24.

The WLQ is a questionnaire to assess health-related work productivity loss. Patients who worked full or part time or volunteered were asked to rate their level of difficulty or ability to perform specific job demands to assess health-related work productivity loss through questionnaire on a scale of 0 (limited none of the time) to 100 (limited all of the time). Then the scale scores were converted into a Productivity Loss Score indicating an estimated percentage of at-work productivity loss due to health. At baseline, the median WLQ productivity loss score in the SIMPONI IV-treated group and the placebo-treated group was 8.8 and 8.9, respectively. The median decrease (improvement) from baseline in WLQ productivity loss score in the SIMPONI IV and placebo groups at Week 14 and 24 were -2.9 vs. -0.45 ($p < 0.001$), and at Week 24 -3.2 vs. -0.7 ($p < 0.001$).

Patients were asked to indicate how much their disease impacted their daily productivity using a 0 to 10 VAS scale (no impact on productivity (0) to high impact on productivity (10)). At baseline, the median impact of disease on daily productivity (VAS) in the SIMPONI IV -treated group and the placebo-treated group score was 6.4. The median decrease (improvement) from baseline in impact of disease on daily productivity (VAS) score in the SIMPONI IV and placebo groups at Week 14 and Week 24 was -3.4 vs. -0.45 ($p < 0.001$) and -3.3 vs. -0.6 ($p < 0.001$), respectively.

Axial spondyloarthritis

Ankylosing spondylitis

The safety and efficacy of SIMPONI were evaluated in a multi-centre, randomised, double-blind, placebo-controlled study (GO-RAISE) in 356 adult patients with active ankylosing spondylitis (defined as a Bath Ankylosing Spondylitis Disease Activity Index (BASDAI) ≥ 4 and a VAS for total back pain of ≥ 4 , on a scale of 0 to 10 cm). Patients enrolled in this study had active disease despite current or previous NSAID or DMARD therapy and had not previously been treated with anti-TNF therapy. SIMPONI or placebo were administered subcutaneously every 4 weeks. Patients were randomly assigned to placebo, SIMPONI 50 mg and SIMPONI 100 mg and were allowed to continue concomitant DMARD therapy

(MTX, SSZ and/or HCQ). The primary endpoint was the percentage of patients achieving Ankylosing Spondylitis Assessment Study Group (ASAS) 20 response at week 14. Placebo-controlled efficacy data were collected and analysed through week 24.

Key results for the 50 mg dose are shown in Table 22 and described below. In general, no clinically meaningful differences in measures of efficacy were observed between the SIMPONI 50 mg and 100 mg dosing regimens through week 24. By study design, patients in the long-term extension may have switched between the 50 mg and 100 mg SIMPONI doses at the discretion of the study physician.

Table 22: Key efficacy outcomes from GO-RAISE

	Placebo	SIMPONI 50 mg*
n ^a	78	138
Responders, % of patients		
ASAS 20		
Week 14	22 %	59 %
Week 24	23 %	56 %
ASAS 40		
Week 14	15 %	45 %
Week 24	15 %	44 %
ASAS 5/6		
Week 14	8 %	50 %
Week 24	13 %	49 %
*: $p \leq 0.001$ for all comparisons		
^a : n reflects randomised patients; actual number of patients evaluable for each endpoint may vary by timepoint		

Among patients remaining in the study and treated with SIMPONI, the proportion of patients with an ASAS 20 and ASAS 40 response were similar from week 24 through week 256.

Statistically significant responses in BASDAI 50, 70 and 90 ($p \leq 0.017$) were also seen at weeks 14 and 24. Improvements in key measures of disease activity were observed at the first assessment (week 4) after the initial SIMPONI administration and were maintained through week 24. Among patients remaining in the study and treated with SIMPONI, similar rates of change from baseline in BASDAI were observed from week 24 through week 256. Consistent efficacy was seen in patients regardless of use of DMARDs (MTX, sulfasalazine and/or hydroxychloroquine), HLA-B27 antigen status or baseline CRP levels as assessed by ASAS 20 responses at week 14.

SIMPONI treatment resulted in significant improvements in physical function as assessed by changes from baseline in BASFI at weeks 14 and 24. Health-related quality of life as measured by the physical component score of the SF-36 was also improved significantly at weeks 14 and 24. Among patients remaining in the study and treated with SIMPONI, improvements in physical function and health-related quality of life were similar from week 24 through week 256.

IV AS study

The efficacy and safety of SIMPONI IV were evaluated in a multicenter, randomized, double-blind, placebo-controlled trial (GO-ALIVE) in 208 adults with active ankylosing spondylitis and inadequate response or intolerance to NSAIDs. Patients had a diagnosis of definite AS for at least 3 months according to modified New York criteria. Patients had symptoms of active disease (Bath AS Disease Activity Index

[BASDAI] ≥ 4 , VAS for total back pain of ≥ 4 , on scales of 0 to 10 cm (0 to 100 mm), and a CRP level of ≥ 0.3 mg/dL (3 mg/L)). Patients were randomized to receive SIMPONI IV 2 mg/kg (N=105) or placebo (N=103) as a 30-minute intravenous infusion at Weeks 0, 4 and 12. All patients on placebo received SIMPONI IV at Week 16, Week 20 and every 8 weeks thereafter through Week 52. Patients in the SIMPONI IV-treatment group continued to receive SIMPONI IV infusions at Week 20 and every 8 weeks through Week 52. Patients were allowed to continue stable doses of concomitant MTX, SSZ, hydroxychloroquine (HCQ), low dose oral corticosteroids (equivalent to ≤ 10 mg of prednisone per day), and/or NSAIDs during the trial. At study enrollment, the use of other DMARDs including cytotoxic agents or other biologics was prohibited.

The median duration of AS disease was 2.8 years, median duration of inflammatory back pain was 8 years, 90% were HLA-B27 positive, 8.2% had prior joint surgery or procedure, 5.8% had complete ankylosis of the spine, 14% had received prior therapy with one TNF blocker (other than golimumab) and discontinued for reasons other than lack of efficacy within the first 16 weeks of treatment (primary failure), and 76% received at least one DMARD in the past. During the trial, the use of concomitant medications was NSAIDs (88%), SSZ (38%), corticosteroids (26%), MTX (18%), and HCQ (0.5%).

The primary endpoint was the percentage of patients achieving an Assessment in Ankylosing Spondylitis (ASAS) 20 response at Week 16. The major secondary endpoints were the proportion of subjects who achieve an ASAS 40 response at Week 16, the proportion of subjects who achieve at least a 50% improvement from baseline in BASDAI at Week 16, and the change from baseline in BASFI at Week 16.

Reduction in signs and symptoms

IV AS study

Treatment with SIMPONI IV, compared with placebo, resulted in a significant improvement in signs and symptoms as demonstrated by the percentage of patients with an ASAS 20 response at Week 16 (see Table 23). In patients treated with SIMPONI IV, 73% of patients achieved ASAS 20 response compared with 26% treated with placebo ($p<0.001$) at Week 16. In addition, ASAS 40 response was achieved at Week 16 by more patients treated with SIMPONI IV (48%) compared with the placebo group (8.7%; $p<0.001$). At Week 16, a greater percentage of patients treated with SIMPONI IV achieved a low level of disease activity (i.e., ASAS partial remission, defined as a value < 2 on a scale of 0-10 cm in each of the four ASAS 20 response parameters) compared with patients treated with placebo (16% vs. 3.9%, $p=0.003$). Compared with placebo, patients treated with SIMPONI IV achieved significantly higher responses as measured by ASAS 5/6 response (65% vs. 12%, respectively, $p<0.001$).

Table 23: IV AS Study: Percentage of ASAS Responders at Week 16: randomized patients

	Placebo N ^a =103	SIMPONI IV N ^a =105	Treatment Difference (95% CI)
% Responders			
ASAS 20	26%	73%	47%^b (35.2, 59.0)
ASAS 40	8.7%	48%	39% ^b (27.9, 49.8)
ASAS partial remission	3.9%	16%	12.3% ^c (4.4, 20.3)
ASAS 5/6	12%	65%	53.0% ^b (42.1, 64.0)

^a N reflects randomized patients.

Bold text indicates primary endpoint.

^b $p<0.001$

^c $p=0.003$

The percentage of patients achieving ASAS 20 responses by visit through Week 16 for IV AS Study is shown in Figure 3. ASAS 20 responses were observed in 37% of patients treated with SIMPONI IV at the first assessment (Week 2) compared with 19% placebo-treated patients.

Figure 3: IV AS Study: Percentage of Patients Achieving an ASAS 20 Response Through Week 16: randomized patients

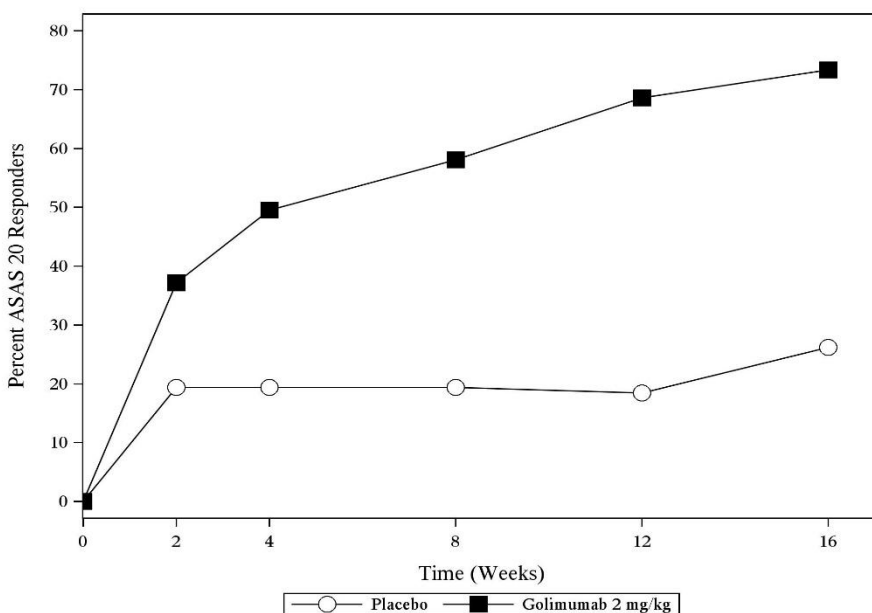


Table 24 shows the improvements in the components of the ASAS response criteria and other measures of disease activity at Week 16 for the SIMPONI IV and placebo groups.

Table 24: IV AS Study: Change from baseline in ASAS Components and Other Measures of Disease Activity at Week 16: randomized patients

	Placebo N=103 ^a			SIMPONI IV N=105 ^a		
	Baseline	Week 16 change baseline	from	Baseline	Week 16 change baseline	from
ASAS 20 Response criteria						
Patient Global Assessment of Disease Activity (0-100 mm) ^b	73	-4.0		73	-30	
Total back pain (0-100 mm) ^c	74	-6.0		72	-32	
BASFI (0-10) ^d	6.6	-0.3		6.5	-2.2*	
Inflammation (0-10) ^e	7.5	-0.8		7.1	-3.3	
Night back pain (0-100 mm) ^c	73	-3.5		73	-33	
CRP (mg/L)	15	-1.4		15	-13	

Note: All values are medians.

* p<0.001

a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary.

b Measured on a Visual Analog Scale (VAS) with 0= very well, 100=very poor

c Measured on a Visual Analog Scale (VAS) with 0= no pain, 100=most severe pain

d BASFI is Bath Ankylosing Spondylitis Functional Index.

e Inflammation is the mean of 2 morning stiffness self-assessments in the BASDAI.

BASDAI is a patient self-assessment using a visual analogue scale (0 to 10 cm) on the following criteria: fatigue, spinal pain, joint pain, enthesitis, qualitative morning stiffness, and quantitative morning stiffness. Patients treated with SIMPONI IV achieved significantly higher responses as measured by a 50%, 70% and 90% improvement from baseline in BASDAI at Week 16 (Table 25). The median baseline BASDAI score, assessed on a VAS scale (0-10 cm), was 7.1 for the SIMPONI IV-treated group and 7.0 for the placebo group. The median decrease (improvement) from baseline in BASDAI score was greater in SIMPONI IV-treated patients compared with placebo-treated patients at Week 16 (-3.0 vs. -0.7, respectively). The percentage of patients achieving a BASDAI 50 response at Week 16 was 41% for patients treated with SIMPONI IV and 15% for patients treated with placebo (p<0.001). At Week 16, a greater proportion of patients treated with SIMPONI IV achieved a BASDAI score of less than 3 compared to patients treated with placebo (38% vs. 13%, respectively).

Table 25: IV AS Study: Percentage of Patients with BASDAI Response at Week 16, randomized patients

	Placebo N ^a =103	SIMPONI IV N ^a =105	Treatment Difference (95% CI)
Responders, % of patients			
BASDAI 50 ^b	15%	41%	26.4* (14.8, 37.9)
BASDAI 70 ^b	5.8%	25%	19.0 (9.6, 28.4)
BASDAI 90 ^b	2.9%	11%	8.5 (1.6, 15.3)
BASDAI Score <3 ^b	13%	38%	25.5 (14.2, 36.7)

* p<0.001

a N reflects randomized patients.

b BASDAI: Patient self-assessment using a VAS (0 to 10 cm) on the following criteria: Fatigue, spinal pain, joint pain, enthesitis, qualitative morning stiffness, and quantitative morning stiffness. BASDAI 50, BASDAI 70 and BASDAI 90 is the proportions of subjects achieving ≥50%, ≥70%, or ≥90% improvement from baseline in BASDAI score.

Patients with enthesitis at baseline (83%) were evaluated for improvement in enthesitis using the UCSF Enthesitis Index on a scale of 0-17. At baseline, the median UCSF Enthesitis Index score in the SIMPONI IV-treated group and the placebo-treated group was 5.0 and 6.0, respectively. SIMPONI IV-treated patients showed an improvement in enthesitis score of 3.0 as compared with an improvement in placebo-treated patients of 1.0 at Week 16 (p<0.001).

Ankylosing Spondylitis Disease Activity Score (ASDAS) is a measure of disease activity and is derived from assessments of total back pain, morning stiffness, patient global assessment, peripheral pain/swelling and measurement of CRP. Major improvement in ASDAS is defined as a decrease ≥ 2.0 and inactive disease defined as score of <1.3. In patients treated with SIMPONI IV, 52% of patients achieved ASDAS major improvement compared with 2.9% treated with placebo (p<0.001) at Week 16. At Week 16, a greater

percentage of patients treated with SIMPONI IV achieved inactive disease compared with patients treated with placebo (20% vs. 2.9%, $p<0.001$).

Improvement in physical function

IV AS study

Bath Ankylosing Spondylitis Functional Index (BASFI) is average of 10 self-assessed questions related to patient's functional anatomy and ability to cope with everyday life. A decreasing score is indicative of improvement in BASFI. The median baseline BASFI score was 6.3 and 6.1 for the SIMPONI IV and placebo groups, respectively. Treatment with SIMPONI IV resulted in significant improvements in physical function as assessed by changes from baseline in BASFI over time and the percentage of patients who achieved a 2-unit improvement in BASFI at Week 16. At Weeks 2 and 16, the respective median decrease (improvement) from baseline in the BASFI for the golimumab and placebo groups were -0.9 and -0.3 at Week 2 and -2.2 vs -0.3 at Week 16. The improvement in physical function was maintained at Week 28 in patients treated with SIMPONI IV. Among patients with a BASFI score ≥ 2 at baseline, a greater percentage of patients treated with SIMPONI IV achieved a 2-unit improvement in BASFI compared with patients treated with placebo (59% vs. 22%, respectively) at Week 16 (see Table 26).

Table 26: IV AS Study: Change from baseline in BASFI analyses; randomized patients

	Placebo N=103 ^a	SIMPONI IV N=105 ^a
BASFI (0-10): median change from baseline ^b		
Baseline (median)	6.6	6.5
Week 16	-0.3	-2.2
> 2-unit improvement in BASFI from baseline ^b: % of patients		
Week 16	22%	59%

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

^b Bath Ankylosing Spondylitis Functional Index (BASFI), average of 10 self-assessed questions related to patient's functional anatomy and ability to cope with everyday life. A negative/decreasing score is indicative of improvement in BASFI.

Improvement in range of motion

IV AS study

Bath Ankylosing Spondylitis Metrology Index (BASMI) is a musculoskeletal assessment and is represented as an aggregate score of 5 components (lumbar flexion, lumbar side flexion, intermalleolar distance, tragus to wall distance, and cervical rotation). At Week 16, the median decrease (improvement) from baseline in BASMI was of greater magnitude in patients treated with SIMPONI IV (-0.7) compared to placebo (-0.1, $p<0.001$). The percentage of patients with at least 1-unit improvement in BASMI at Week 16 in patients treated with SIMPONI IV (38%) was greater compared with patients treated with placebo (9.9%, $p<0.001$) (see Table 27).

Table 27: IV AS Study: Median change from baseline in BASMI component scores and other measures of spinal mobility; randomized patients

	Placebo (N=103) ^a	SIMPONI IV (N=105) ^a	p-value
BASMI (0-10): median change from baseline			
Baseline (median)	4.6	4.5	
Week 16	-0.1	-0.7	<0.001
≥ 1-unit improvement in BASMI from baseline: % of patients			
Week 16	9.9%	38.0%	<0.001

Lumbar flexion (modified Schober's): median change from baseline^b			
Baseline (median)	6.3	6.3	
Week 16	0.0	-0.4	0.026
Lumbar side flexion: median change from baseline^b			
Baseline (median)	6.0	6.1	
Week 16	0.00	-0.5	<0.001
Intermalleolar distance: median change from baseline^b			
Baseline (median)	3.5	3.5	
Week 16	-0.3	-0.6	0.047
Tragus-to-wall distance: median change from baseline^b			
Baseline (median)	2.1	2.1	
Week 16	0.0	-0.1	NS
Cervical rotation: median change from baseline^b			
Baseline (median)	5.9	5.2	
Week 16	-0.1	-0.7	<0.001
Chest expansion (cm): median change from baseline			
Baseline (median)	3.0	3.0	
Week 16	0.0	0.8	NS

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

^b BASMI components values are presented as S scores, using the van der Heijde calculation.

Improvement in health-related quality of life

IV AS study

Health-related quality of life was measured by the SF 36 (Short Form 36 Health Survey) Physical and Mental Component Summaries, ASQoL (Ankylosing Spondylitis Quality of Life questionnaire), EQ 5D-5L (EuroQoL 5 Dimensions questionnaire) Index Score, and EQ VAS Score at baseline, Week 8, Week 16 and Week 28.

Patients receiving SIMPONI IV demonstrated significantly greater median improvement from baseline compared with placebo in the physical component summary (PCS, 7.8 vs. 2.7, $p<0.001$) score at Week 16. At baseline, the median SF-36 PCS scores of 31 in the SIMPONI IV group and 31 for the placebo group. Greater proportions of subjects in the SIMPONI IV group compared with placebo achieved a clinically meaningful change from baseline in SF-36 PCS score (an increase of 5 or more units) at Week 8 (58% vs 27%) and Week 16 (68% vs 36%).

Patients receiving SIMPONI IV demonstrated significantly greater median improvement from baseline compared with placebo in the mental component summary (MCS, 5.7 vs. 1.8, $p<0.001$) score at Week 16. At baseline, the median SF-36 MCS scores were 40 in the SIMPONI IV group and 41 for the placebo group. Greater proportions of subjects in the SIMPONI IV group compared with placebo achieved a clinically meaningful change from baseline in SF-36 MCS score (an increase of 5 or more units) from Week 8 (49% vs 34%) and Week 16 (54% vs 29%).

In addition, the eight domains of the SF-36 that comprise the PCS and MCS scores showed greater median improvements at Week 8 and Week 16 for patients treated with SIMPONI IV compared to the placebo.

Ankylosing Spondylitis Quality of Life Questionnaire (ASQoL) is a disease-specific instrument used to measure quality of life in the AS patient population. It consists of 18 items requesting a yes or no response to questions related to the impact of pain on sleep, mood, motivation, ability to cope, activities of daily living, independence, relationships, and social life. Lower scores indicate improvement. Median baseline

ASQoL scores in the SIMPONI IV-treated group and placebo group were 14 and 13, respectively. Patients treated with SIMPONI IV achieved greater median decrease (improvement) from baseline in the ASQoL score compared to placebo-treated patients at Week 16 (-4.0 vs. -1.0, $p < 0.001$).

EuroQoL 5D-5L is a measure of health status to provide a measure of health for clinical and economic appraisal. EQ-5D-5L consists of 2 elements: The EQ-5D-5L descriptive system and the EQ visual analogue scale (EQ VAS). The EQ-5D descriptive system comprises the following 5 dimensions: mobility, self-care, usual activities, pain/discomfort and anxiety/depression. EQ VAS records the respondent's self-rated health on a VAS, with the endpoints of "worst imaginable health" and "best imaginable health".

Patients treated with SIMPONI achieved median improvement from baseline in the EQ-5D-5L Index Score and EQ VAS Score compared to placebo-treated patients at Week 16 (0.2 vs. 0.05 and 20 mm vs. 5.0 mm, respectively). Median baseline values for EQ-5D-5L index score and EQ VAS score in the SIMPONI IV-treated group and placebo group were 0.5 vs. 0.6 and 37 mm vs. 37 mm, respectively (see Table 28).

Table 28: IV AS Study: Health-related quality of life analyses at Week 16; randomized and treated patients

	Placebo (N=103)^a	SIMPONI IV (N=105)^a
SF-36 PCS		
Baseline score		
Mean $\pm$ SD	32 $\pm$ 5.9	32 $\pm$ 5.6
Median	31	31
Week 16, improvement from baseline		
Mean $\pm$ SD	2.9 $\pm$ 6.2)	8.5 $\pm$ 7.5
Median	2.7	7.8
SF 36 MCS		
Baseline score		
Mean $\pm$ SD	42 $\pm$ 10.2	40 $\pm$ 10.4
Median	41	40
Week 16, improvement from baseline		
Mean $\pm$ SD	0.8 $\pm$ 10.0	6.5 $\pm$ 9.1
Median	1.8	5.7
ASQoL Score		
Baseline Score		
Mean $\pm$ SD	12 $\pm$ 4.1	13 $\pm$ 4.0
Median	13	14
Week 16, change from baseline ^b		
Mean $\pm$ SD	-1.8 $\pm$ 4.6	-5.4 $\pm$ 5.0
Median	-1.0	-4.0
EQ 5D Index^c		
Baseline Score ^c		
Mean $\pm$ SD	0.6 $\pm$ 0.15	0.5 $\pm$ 0.16
Median	0.6	0.5
Week 16, improvement from baseline		
Mean $\pm$ SD	0.05 $\pm$ 0.14	0.2 $\pm$ 0.16

Median	0.05	0.2
EQ 5D VAS^c		
Baseline Score (mm)		
Mean $\pm$ SD	41 $\pm$ 18.1	42 $\pm$ 22.7
Median	37	37
Week 16, (mm), improvement from baseline		
Mean $\pm$ SD	4.8 $\pm$ 23.4	20 $\pm$ 24.6
Median	5.0	20

^a N reflects randomized patients; actual number of patients evaluable for each endpoint may vary by timepoint.

^b A negative/decreasing score is indicative of improvement.

^c EuroQoL 5D-5L is a measure of health status to provide a measure of health for clinical and economic appraisal. EQ-5D-5L consists of 2 elements: The EQ-5D-5L descriptive system and the EQ visual analogue scale (EQ VAS). The EQ-5D descriptive system comprises the following 5 dimensions: mobility, self-care, usual activities, pain/discomfort and anxiety/depression. EQ VAS records the respondent's self-rated health on a VAS, with the endpoints of "worst imaginable health" and "Best imaginable health".

Health economics

IV AS study

Health economics data on patient productivity at work was collected through the Work Limitations Questionnaire (WLQ) and the productivity visual analog scale (VAS) at baseline and Weeks 8, 16 and 28.

The WLQ is a questionnaire to assess health-related work productivity loss. Patients who worked full or part time or volunteered were asked to rate their level of difficulty or ability to perform specific job demands to assess health-related work productivity loss through questionnaire on a scale of 0 (limited none of the time) to 100 (limited all of the time). Then the scale scores were converted in to a Productivity Loss Score indicating an estimated percentage of at-work productivity loss due to health. At baseline, the median WLQ productivity loss scores in the SIMPONI IV-treated group and placebo group were 11 and 11, respectively. The median decrease (improvement) from baseline in WLQ productivity loss score in the SIMPONI IV and placebo groups at Week 16 was -2.8 vs. -0.7.

Patients were asked to indicate how much their disease affected their productivity using a 0 to 10 VAS scale (no impact on productivity (0) to high impact on productivity (10)). At baseline, the median impact of disease on daily productivity (VAS) scores in the SIMPONI IV-treated group and placebo group were 7.6 and 7.3. The median decrease (improvement) from baseline in impact of disease on daily productivity (VAS) score in the SIMPONI IV and placebo groups at Week 16 was -2.6 vs. -0.7.

Improvement in sleep

IV AS study

In the IV AS study, the extent of sleep problems was assessed using the Medical Outcomes Study Sleep Scale (MOS-SS), which measured 6 dimensions of sleep, including sleep disturbance, somnolence, sleep adequacy, snoring, shortness of breath or headache, and quantity of sleep/optimal sleep. At baseline, the median sleep problems index score was 39 in the SIMPONI IV and placebo groups. At Week 16, median improvement in sleep problems index was greater for patients treated with SIMPONI IV compared with the placebo group (7.0 vs. 2.8). An increase from baseline represents improvement (Table 29).

Table 29: IV AS Study: Sleep Problems Index Score (MOS-SS); randomized patients

	Placebo (N=103) ^a	SIMPONI IV (N=105) ^a
Composite Baseline score		
Mean $\pm$ SD	39 (8.3)	40 (7.8)
Median	39	39
Improvement at Week 16		
Mean $\pm$ SD	2.5 (8.16)	6.6 (7.18)
Median	2.8	7.0

^a N reflects randomized patients; however, N may vary at different time points.

Medical Outcomes Study Sleep Scale (MOS-SS) measures 6 dimensions of sleep, including sleep disturbance, somnolence, sleep adequacy, snoring, shortness of breath or headache, and quantity of sleep/optimal sleep.

Non-radiographic axial spondyloarthritis

The safety and efficacy of SIMPONI were evaluated in a multi-centre, randomised, double-blind, placebo-controlled study (GO-AHEAD) in 197 adult patients with severe active nr-Axial SpA (defined as those patients meeting the ASAS classification criteria of axial spondyloarthritis but did not meet the modified New York criteria for AS). Patients enrolled in this study had active disease (defined as a BASDAI ≥ 4 and a Visual Analogue Scale (VAS) for total back pain of ≥ 4 , each on a scale of 0-10 cm) despite current or previous NSAID therapy and had not previously been treated with any biological agents including anti-TNF therapy. Patients were randomly assigned to placebo or SIMPONI 50 mg administered subcutaneously every 4 weeks. At week 16, patients entered an open label period in which all patients received SIMPONI 50 mg administered subcutaneously every 4 weeks through week 48 with efficacy assessments performed through week 52 and safety follow-up through week 60.

Approximately 93% of patients who were receiving SIMPONI at the beginning of the open-label extension (week 16) remained on treatment through the end of the study (week 52). Analyses were performed on both the All Treated (AT, N = 197) and Objective Signs of Inflammation (OSI, N = 158, defined by elevated CRP and/or evidence of sacroiliitis on MRI at baseline) populations. Placebo-controlled efficacy data were collected and analysed through week 16. The primary endpoint was the proportion of patients achieving ASAS 20 response at week 16. Key results are shown in Table 30 and described below.

Table 30
Key efficacy outcomes from GO-AHEAD at week 16

Improvements in signs and symptoms				
	All treated population (AT)		Objective signs of inflammation population (OSI)	
	Placebo	SIMPONI 50 mg	Placebo	SIMPONI 50 mg
n ^a	100	97	80	78
Responders, % of patients				
ASAS 20	40%	71%**	38%	77%**
ASAS 40	23%	57%**	23%	60%**
ASAS 5/6	23%	54%**	23%	63%**
ASAS Partial Remission	18%	33%*	19%	35%*
ASDAS-C ^b < 1.3	13%	33%*	16%	35%*
BASDAI 50	30%	58%**	29%	59%**
Inhibition of inflammation in sacroiliac (SI) joints as measured by MRI				
	Placebo	SIMPONI 50 mg	Placebo	SIMPONI 50 mg
n ^c	87	74	69	61
Mean change in SPARCC ^d MRI sacroiliac joint score	-0.9	-5.3**	-1.2	-6.4**

- a n reflects randomised and treated patients
- b Ankylosing Spondylitis Disease Activity Score C-Reactive Protein (AT-Placebo, N = 90; AT-SIMPONI 50 mg, N = 88; OSI-Placebo, N = 71; OSI-SIMPONI 50 mg, N = 71)
- c n reflects number of patients with baseline and week 16 MRI data
- d SPARCC (Spondyloarthritis Research Consortium of Canada)
- ** $p < 0.0001$ for SIMPONI vs placebo comparisons
- * $p < 0.05$ for SIMPONI vs placebo comparisons

Statistically significant improvements in signs and symptoms of severe active nr-Axial SpA were demonstrated in patients treated with SIMPONI 50 mg compared to placebo at week 16 (Table 30). Improvements were observed at the first assessment (week 4) after the initial SIMPONI administration. SPARCC score as measured by MRI showed statistically significant reductions in SI joint inflammation at week 16 in patients treated with SIMPONI 50 mg compared to placebo (Table 30). Pain as assessed by the Total Back Pain and Nocturnal Back Pain VAS, and disease activity as measured by ASDAS-C also showed statistically significant improvement from baseline to week 16 in patients treated with SIMPONI 50 mg compared to placebo ($p < 0.0001$).

Statistically significant improvements in spinal mobility as assessed by BASMI (Bath Ankylosing Spondylitis Metrology Index) and in physical function as assessed by the BASFI were demonstrated in SIMPONI 50 mg-treated patients as compared to placebo-treated patients ($p < 0.0001$). Patients treated with SIMPONI experienced significantly more improvements in health-related quality of life as assessed by ASQoL, EQ-5D, and physical and mental components of SF-36, and experienced significantly more improvements in productivity as assessed by greater reductions in overall work impairment and in activity impairment as assessed by the WPAI questionnaire than patients receiving placebo.

For all of the endpoints described above, statistically significant results were also demonstrated in the OSI population at week 16.

In both the AT and OSI populations, the improvements in signs and symptoms, spinal mobility, physical function, quality of life, and productivity observed at week 16 among patients treated with SIMPONI 50 mg continued in those remaining in the study at week 52.

Ulcerative colitis:

The efficacy of SIMPONI was evaluated in two randomized, double-blind, placebo-controlled clinical studies in adult patients.

The induction study (PURSUIT-Induction) evaluated patients with moderately to severely active ulcerative colitis (Mayo score 6 to 12; Endoscopy subscore ≥ 2) who had an inadequate response to or failed to tolerate conventional therapies, or were corticosteroid dependent. In the dose confirming portion of the study, 761 patients were randomized to receive either 400 mg SIMPONI SC at week 0 and 200 mg at week 2, 200 mg SIMPONI SC at week 0 and 100 mg at week 2, or placebo SC at weeks 0 and 2. Concomitant stable doses of oral aminosalicylates, corticosteroids, and/or immunomodulatory agents were permitted. The efficacy of SIMPONI through week 6 was assessed in this study.

The results of the maintenance study (PURSUIT-Maintenance) were based on evaluation of 456 patients who achieved clinical response from previous induction with SIMPONI. Patients were randomized to receive SIMPONI 50 mg, SIMPONI 100 mg or placebo administered subcutaneously every 4 weeks. Concomitant stable doses of oral aminosalicylates, and/or immunomodulatory agents were permitted. Corticosteroids were to be tapered at the start of the maintenance study. The efficacy of SIMPONI through week 54 was assessed in this study. Patients who completed the maintenance study through week 54

continued treatment in a study extension, with efficacy evaluated through week 216. Efficacy evaluation in the study extension was based on changes in corticosteroid use, Physician's Global Assessment (PGA) of disease activity, and improvement in quality of life as measured by Inflammatory Bowel Disease Questionnaire (IBDQ).

Table 31: Key efficacy outcomes from PURSUIT - Induction and PURSUIT – Maintenance

PURSUIT-Induction			
	Placebo N = 251	SIMPONI 200/100 mg N = 253	
Percentage of patients			
Patients in clinical response at week 6 ^a	30%	51%**	
Patients in clinical remission at week 6 ^b	6%	18%**	
Patients with mucosal healing at week 6 ^c	29%	42%*	
PURSUIT-Maintenance			
	Placebo ^d N = 154	SIMPONI 50 mg N = 151	SIMPONI 100 mg N = 151
Percentage of patients			
Maintenance of response (Patients in clinical response through week 54) ^e	31%	47%*	50%**
Sustained remission (Patients in clinical remission at both week 30 and week 54) ^f	16%	23% ^g	28%*
N = number of patients ** p ≤ 0.001 * p ≤ 0.01 ^a defined as a decrease from baseline in the Mayo score by ≥ 30% and ≥ 3 points, accompanied by a decrease in the rectal bleeding subscore of ≥ 1 or a rectal bleeding subscore of 0 or 1. ^b Defined as a Mayo score ≤ 2 points, with no individual subscore > 1. ^c Defined as 0 or 1 on the endoscopy subscore of the Mayo score. ^d SIMPONI induction only. ^e Patients were assessed for UC disease activity by partial Mayo score every 4 weeks (loss of response was confirmed by endoscopy). Therefore, a patient who maintained response was in a state of continuous clinical response at each evaluation through week 54. ^f A patient had to be in remission at both weeks 30 and 54 (without demonstrating a loss of response at any time point through week 54) to achieve durable remission. ^g In patients weighing less than 80 kg, a greater proportion of patients who received 50 mg maintenance therapy showed sustained clinical remission compared with those who received placebo.			

More SIMPONI-treated patients demonstrated sustained mucosal healing (patients with mucosal healing at both week 30 and week 54) in the 50 mg group (42%, nominal $p < 0.05$) and 100 mg group (42%, $p < 0.005$) compared with patients in the placebo group (27%).

Among the 54% of patients (247/456) who were receiving concomitant corticosteroids at the start of PURSUIT-Maintenance, the proportion of patients who maintained clinical response through week 54 and were not receiving concomitant corticosteroids at week 54 was greater in the 50 mg group (38%, 30/78) and 100 mg group (30%, 25/82) compared with the placebo group (21%, 18/87). The proportion of patients who eliminated corticosteroids by week 54 was greater in the 50 mg group (41%, 32/78) and 100 mg group (33%, 27/82) compared with the placebo group (22%, 19/87). Among patients who entered the study extension, the proportion of subjects who remained corticosteroid free was generally maintained through week 216.

Patients who did not achieve clinical response at week 6 in the PURSUIT-Induction studies were dosed Simponi 100 mg every 4 weeks in the PURSUIT-Maintenance study. At week 14, 28% of these patients achieved response defined by partial Mayo score (decreased by ≥ 3 points compared with start of induction). At week 54, the clinical outcomes observed in these patients were similar to the clinical outcomes reported for the patients achieving clinical response at week 6.

At week 6, SIMPONI significantly improved quality of life as measured by change from baseline in a disease specific measure, IBDQ (inflammatory bowel disease questionnaire). Among patients who received SIMPONI maintenance treatment, the improvement in quality of life as measured by IBDQ was maintained through week 54.

Approximately 63% of patients who were receiving Simponi at the beginning of the study extension (week 56), remained on treatment through the end of the study (last golimumab administration at week 212).

Immunogenicity

Across the Phase III RA, PsA and AS studies through week 52, antibodies to golimumab were detected by the enzyme immunoassay (EIA) method in 5 % (105/2062) of golimumab treated patients and, where tested, nearly all antibodies were neutralising *in vitro*. Similar rates were shown across rheumatologic indications. Treatment with concomitant MTX resulted in a lower proportion of patients with antibodies to golimumab than patients receiving golimumab without MTX (approximately 3% [41/1235] versus 8% [64/827], respectively).

In nr-Axial SpA, antibodies to golimumab were detected in 7% (14/193) of golimumab treated patients through week 52 by the EIA method.

In the Phase II and III UC studies through week 54, antibodies to golimumab were detected by the EIA method in 3% (26/946) of golimumab treated patients. Sixty-eight percent (21/31) of antibody-positive patients had neutralizing antibodies *in vitro*. Treatment with concomitant immunomodulators (azathioprine, 6-mercaptopurine and MTX) resulted in a lower proportion of patients with antibodies to golimumab than patients receiving golimumab without immunomodulators (1% (4/308) versus 3% (22/638), respectively). Of patients that continued in the study extension and had evaluable samples through week 228, antibodies to golimumab were detected in 4% (23/604) of golimumab treated patients. Eighty-two percent (18/22) of antibody-positive patients had neutralizing antibodies *in vitro*.

The presence of antibodies to golimumab may increase the risk of injection site reactions (see *Warnings and Precautions*). The small number of patients positive for antibodies to golimumab limits the ability to draw definitive conclusions regarding the relationship between antibodies to golimumab and clinical efficacy or safety measures.

Because immunogenicity analyses are product- and assay-specific, comparison of antibody rates with those from other products is not appropriate.

Following IV administration of SIMPONI in combination with MTX in RA patients, antibodies to golimumab were detected in 4.2% (39/922) of golimumab-treated patients through approximately 1 year. All patients who were positive for antibodies to golimumab had neutralizing antibodies *in vitro*.

The small number of patients positive for antibodies to golimumab limits the ability to draw definitive conclusions regarding the relationship between antibodies to golimumab and clinical efficacy or safety measures.

The data reflect the percentage of patients whose test results were considered positive for antibodies to golimumab in an ELISA assay, and are highly dependent on the sensitivity and specificity of the assay. Additionally, the observed incidence of antibody positivity in an assay may be influenced by several factors including sample handling, timing of sample collection, concomitant medications, and underlying disease. For these reasons, comparison of the incidence of antibodies to golimumab with the incidence of antibodies to other products may be misleading.

Following IV administration in patients with RA, PsA or AS, antibodies to golimumab were detected by the drug-tolerant EIA method in 20% of SIMPONI-treated patients (RA: 21%, PsA: 19% and AS: 19%). Where tested, approximately one-third of the antibodies to golimumab were neutralizing. Treatment with concomitant MTX resulted in a slightly lower proportion of patients with antibodies to golimumab than patients receiving SIMPONI without MTX (approximately 19% vs. 25%, respectively).

The higher incidence of antibodies to golimumab with the drug-tolerant EIA method were mostly due to low-titer antibodies, which did not have an apparent impact on drug concentrations, efficacy and safety. Although higher-titer antibodies, which were mostly neutralizing, may be associated with lower drug concentrations and diminished efficacy, there were few patients with high titers in the IV PsA and IV AS studies. Development of antibodies to golimumab did not preclude clinical response.

The observed incidence of antibody positivity may be influenced by several factors including sample handling, timing of sample collection, concomitant medications, underlying disease, and particularly the sensitivity of the assay. For these reasons, comparison of the incidence of antibodies to SIMPONI with the incidence of antibodies to other products or results from different assays may be misleading.

Pharmacokinetic Properties

Absorption

Following a single subcutaneous administration of golimumab to healthy subjects or patients with RA, the median time to reach maximum serum concentrations (T_{max}) ranged from 2 to 6 days. A subcutaneous injection of 50 mg golimumab to healthy subjects produced a mean $\pm$ standard deviation maximum serum concentration (C_{max}) of 3.1 ± 1.4 mcg/ml.

Following a single subcutaneous injection of 100 mg, the absorption of golimumab was similar in the upper arm, abdomen, and thigh, with a mean absolute bioavailability of 51%. Since golimumab exhibited approximately dose proportional PK following a subcutaneous administration, the absolute bioavailability of a golimumab 50 mg or 200 mg dose is expected to be similar.

Distribution

Following a single IV administration, the mean volume of distribution was 115 ± 19 mL/kg.

Elimination

The systemic clearance of golimumab was estimated to be 6.9 ± 2.0 mL/day/kg. Terminal half-life value was estimated to be approximately 12 ± 3 days in healthy subjects and similar values were observed in patients with RA, PsA, AS, or UC.

When 50 mg golimumab was administered subcutaneously to patients with RA, PsA or AS every 4 weeks, serum concentrations reached steady state by Week 12. With concomitant use of MTX, treatment with 50 mg golimumab subcutaneously every 4 weeks resulted in a mean ($\pm$ standard deviation) steady-state trough serum concentration of approximately 0.6 ± 0.4 mcg/ml in RA patients with active RA despite MTX therapy, and approximately 0.5 ± 0.4 mcg/ml in patients with active PsA and approximately 0.8 ± 0.4 mcg/ml in patients with AS. Steady-state trough mean serum golimumab concentrations in patients with nr-Axial SpA were similar to those observed in patients with AS following subcutaneous administration of

50 mg golimumab every 4 weeks.

Patients with RA, PsA or AS who did not receive concomitant MTX had approximately 30% lower steady-state trough concentrations of golimumab than those who received golimumab with MTX. In a limited number of RA patients treated with subcutaneous golimumab over a 6-month period, concomitant use of MTX reduced the apparent clearance of golimumab by approximately 36%. However, population pharmacokinetic analysis indicated that concomitant use of NSAIDs, oral corticosteroids or sulfasalazine did not influence the apparent clearance of golimumab.

Following induction doses of 200 mg and 100 mg golimumab at week 0 and 2, respectively, and maintenance doses of 50 mg or 100 mg golimumab subcutaneously every 4 weeks thereafter to patients with UC, serum golimumab concentrations reached steady state approximately 14 weeks after the start of therapy. Treatment with 50 mg or 100 mg golimumab subcutaneous every 4 weeks during maintenance resulted in a mean steady-state trough serum concentration of approximately 0.9 ± 0.5 mcg/mL and 1.8 ± 1.1 mcg/mL, respectively.

In UC patients treated with 50 mg or 100 mg golimumab subcutaneously every 4 weeks, concomitant use of immunomodulators did not have a substantial effect on steady-state trough levels of golimumab.

Patients who developed antibodies to golimumab generally had low trough steady-state serum concentrations of golimumab (see *Pharmacodynamic Properties*).

Linearity

Golimumab exhibited approximately dose-proportional pharmacokinetics in patients with RA over the dose range of 0.1 to 10.0 mg/kg following a single intravenous dose. Following a single SC dose in healthy subjects, approximately dose-proportional pharmacokinetics were also observed over a dose range of 50 mg to 400 mg.

Effect of weight on pharmacokinetics

There was a trend toward higher apparent clearance of golimumab with increasing weight (see *Dosage and Administration*).

Preclinical Safety Data

Non-clinical data reveal no special hazard for humans based on conventional studies of safety pharmacology, repeated dose toxicity, toxicity to reproduction and development.

No mutagenicity studies, animal fertility studies nor long-term carcinogenic studies have been conducted with golimumab.

In a fertility and general reproductive function study in mouse, using an analogous antibody that selectively inhibits the functional activity of mouse TNF- α , the number of pregnant mice was reduced. It is not known whether this finding was due to effects on the males and/or the females. In a developmental toxicity study conducted in mice following administration of the same analogous antibody, and in cynomolgus monkeys using golimumab, there was no indication of maternal toxicity, embryotoxicity or teratogenicity.

PHARMACEUTICAL INFORMATION

List of Excipients

Autoinjector/pre-filled pen:

L-histidine

L-histidine monohydrochloride monohydrate

Polysorbate 80

Sorbitol
Water for injections

Vial

L-histidine
L-histidine monohydrochloride monohydrate
Polysorbate 80
Sorbitol
Water for injection

Incompatibilities

Specific drug compatibility studies have not been conducted.

Shelf Life

Pre-filled pen:

36 months

Vial:

36 months

See expiry date on the outer pack.

Storage Conditions

Store in a refrigerator (2°C – 8°C). Store in original carton until time of use. Protect from light. Do not freeze. Do not shake. Keep out of the sight and reach of children.

SIMPONI (pre-filled pen or vial) may be stored at room temperature up to a maximum of 25°C for a single period of up to 30 days in the original carton; after which, it should not be refrigerated again. SIMPONI must be protected from light. It should be discarded if not used within 30 days of removal from refrigeration.

Vial

Once the solution has been diluted, it should be stored at room temperature, protected from light and the infusion should be completed within 6 hours of preparation.

Nature and Contents of Container

Autoinjector/pre-filled pen:

SIMPONI is supplied as a single-use, sterile solution in a Type 1 glass syringe with a fixed stainless steel needle. The syringe is fitted with a passive safety guard or contained in a single-use autoinjector/pre-filled pen. The syringe is stoppered with a coated stopper and the needle is covered with a needle shield to prevent leakage of the solution through the needle and to protect the needle during handling prior to administration. The fixed needle is a 5 bevel, 27G, half inch stainless steel needle. The needle shields are manufactured using a dry natural rubber containing latex (see *Warnings and Precautions*).

SIMPONI does not contain preservatives. The solution is clear to slightly opalescent, colorless to light yellow with a pH of approximately 5.5. Each mL of SIMPONI contains 100 mg of golimumab, 0.87 mg L-histidine and L-histidine hydrochloride, 41.0 mg sorbitol, 0.15 mg polysorbate 80, and Water for Injection. SIMPONI is available in two strengths: 50 mg of golimumab in 0.5 mL and 100 mg of golimumab in 1 mL.

SIMPONI is available in pack of 1 autoinjector/pre-filled pen.

SIMPONI is packaged in a single use outer carton.

Vial

SIMPONI for intravenous infusion is supplied in single-use vials.

SIMPONI does not contain preservatives. The solution is colorless to light yellow with a pH of approximately 5.5. Each 4 mL vial of SIMPONI contains 50 mg golimumab, 1.14 mg L-histidine and 6.42 mg L-histidine monohydrochloride monohydrate, 180 mg sorbitol, 0.6 mg polysorbate 80, and Water for Injection.

SIMPONI is packaged in a single-use outer carton.

Instructions for Use and Handling and Disposal

Vial

Use aseptic technique.

1. Calculate the dose and the number of SIMPONI vials needed based on patient weight. Each 4 mL vial of SIMPONI contains 50 mg of golimumab.
2. Check that the solution is colorless to light yellow. The solution may develop a few fine translucent particles, as golimumab is a protein. Do not use if opaque particles, discoloration, or other foreign particles are present.
3. Dilute the total volume of the SIMPONI solution dose to 100 mL with 0.9% w/v sodium chloride for infusion. This can be accomplished by withdrawing a volume of the 0.9% w/v sodium chloride solution from the 100-mL glass bottle or infusion bag equal to the volume of SIMPONI and discard the withdrawn solution. Alternatively, SIMPONI can be diluted using the same method described above with 0.45% w/v sodium chloride for infusion.
4. Slowly add the total volume of SIMPONI solution to the 100-mL infusion bottle or bag. Gently mix.
5. Visually inspect parenteral medicinal products for particulate matter or discoloration prior to administration. Do not use if visibly opaque particles, discoloration or foreign particles are observed.
6. Infuse the diluted solution over a period of 30 ± 10 minutes. The infusion of the diluted solution should be completed within 6 hours after preparation.
7. Use only an infusion set with an in-line, sterile, non-pyrogenic, low protein-binding filter (pore size 0.22 micrometer or less). Do not store any unused portion of the infusion solution for reuse.
8. No physical biochemical compatibility studies have been conducted to evaluate the co-administration of SIMPONI with other agents. Do not infuse SIMPONI concomitantly in the same intravenous line with other agents.
9. Any unused product or waste material should be disposed of in accordance with local requirements.

Instructions for injecting SIMPONI under the skin yourself

At the start of your therapy, SIMPONI may be injected by medical or nursing staff. However, your doctor may decide that it is right for you or your caregiver to learn how to inject SIMPONI under the skin

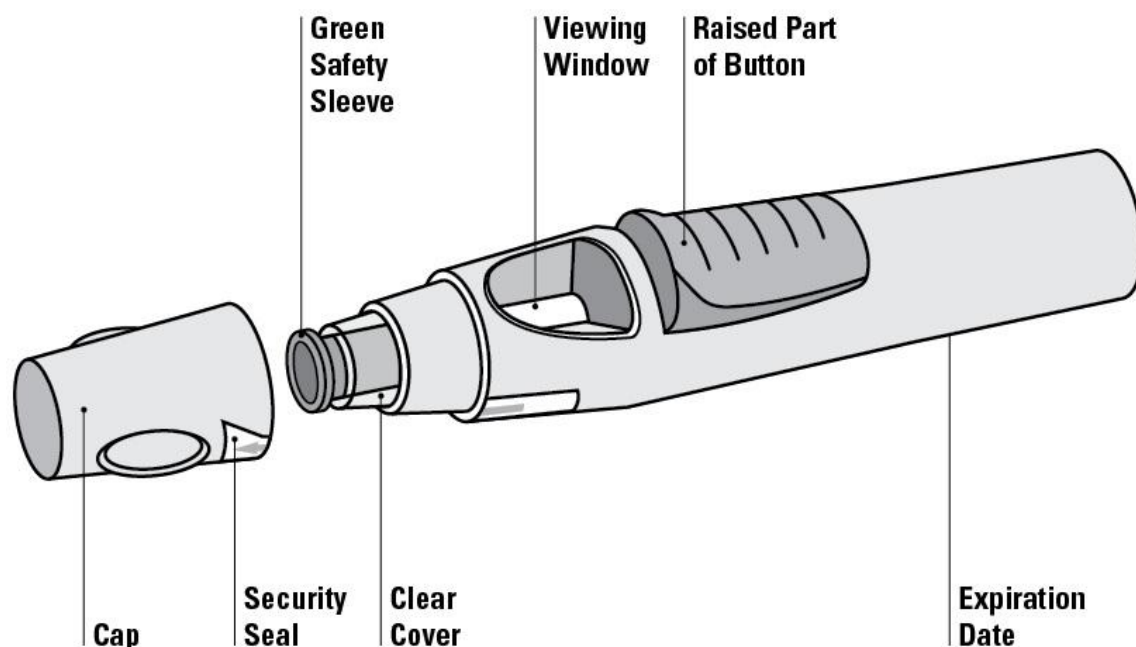
(subcutaneously) yourself. Make sure you (or your caregiver) have been trained how to inject SIMPONI before you do it yourself. Call your doctor if you have any questions about giving yourself an injection.

INSTRUCTIONS FOR INJECTING SIMPONI USING A SINGLE-USE AUTOINJECTOR / PRE-FILLED PEN

If you would like to self-inject SIMPONI, you must be trained by a healthcare professional to prepare an injection and give it to yourself. If you have not been trained, please contact your healthcare professional to schedule a training session.

STEP 1: PREPARING TO USE THE AUTOINJECTOR

The diagram below shows what the autoinjector looks like:



Do **NOT** shake the autoinjector at any time.

Do **NOT** remove the autoinjector cap from the pen until immediately before the injection.

Do **NOT** put the autoinjector cap back on if removed to avoid bending the needle.

Check Expiration Date

- Check the expiration date (indicated as “EXP”) on the autoinjector.
- You can also check the expiration date printed on the carton.
- If the expiration date has passed, **do not** use the autoinjector. Please contact your doctor or pharmacist for assistance.

Check Security Seal

- Check the security seal around the cap of the autoinjector. If the security seal is broken, **do not** use the autoinjector and please contact your doctor or pharmacist for assistance.

Wait 30 Minutes

- To ensure proper injection, allow the autoinjector to sit at room temperature outside the carton for 30 minutes out of the reach of children.



Do **NOT** warm the autoinjector in any other way (for example, **DO NOT** warm it in a microwave or in hot water).

Do **NOT** remove the autoinjector cap while allowing it to reach room temperature.

Assemble Additional Supplies

- Assemble additional supplies you will need for your injection. These include an alcohol swab, a cotton ball or gauze, and a sharps container.

Check the Liquid in the Autoinjector

- Look through the viewing window to make sure that the liquid in the autoinjector is clear to slightly opalescent and colorless to slightly yellow.
- You may also notice an air bubble – this is normal.

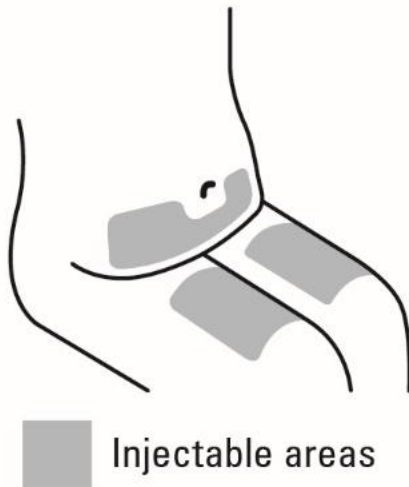
Do **NOT** use if the liquid is discolored, cloudy or contains particles. If this is the case, please contact your doctor or pharmacist for assistance.

STEP 2: CHOOSING AND PREPARING THE INJECTION SITE

Choose the Injection Site

Select from the following areas for your injection:

- Front of the middle thighs.
- Lower abdomen below the belly button, except for the two-inch area directly underneath the belly button.
- Do **NOT** inject into the arm.



DO NOT inject into the arm to avoid the failure of the device and/or unintentional injury.

- If multiple injections are required for a single administration, the injections should be administered at different injection sites on the body.

Do **NOT** inject into areas where the skin is tender, bruised, red, scaly, or hard. Avoid areas with scars or stretch marks.

Wash Hands and Clean Injection Site

- Thoroughly wash your hands with soap and warm water.
- Wipe the injection site with an alcohol swab.

Do **NOT** touch this area again before giving the injection. Allow the skin to dry before injecting.

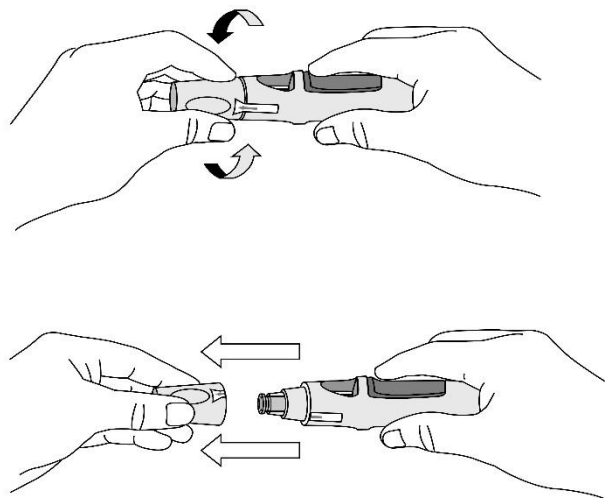
Do **NOT** fan or blow on the clean area.

STEP 3: INJECTING THE MEDICINE

Remove the Cap

The cap should **NOT** be removed until you are ready to inject the medication. The medication should be injected within 5 minutes after the cap has been removed.

- When you are ready to inject, twist the cap slightly to break the security seal.
- Pull the cap off and immediately place the cap into the trash.



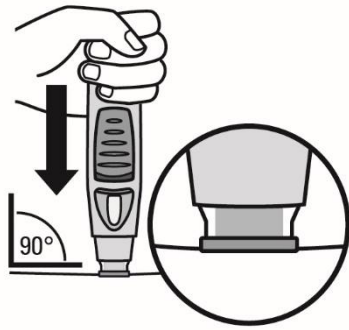
Do **NOT** put the cap back on because it may damage the needle inside the autoinjector.

Note: Do not use autoinjector if it is dropped without the cap in place. If you drop the autoinjector without the cap in place, please contact your doctor or pharmacist for assistance.

Push the Autoinjector Against the Skin Without Pinching the Skin

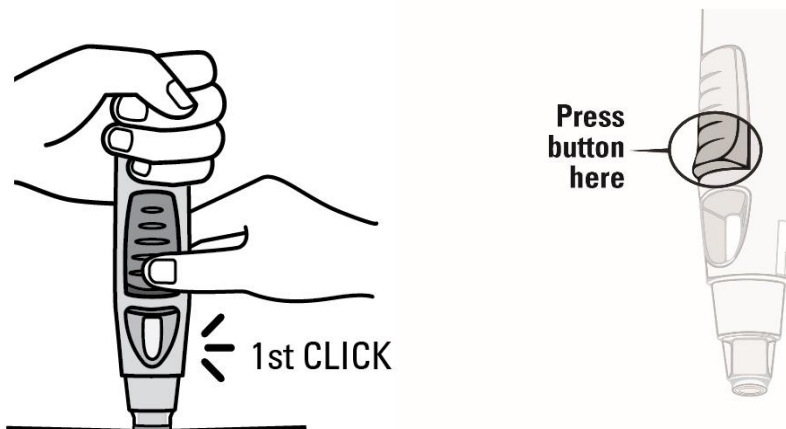


- Hold the autoinjector comfortably with one hand **above the blue button**.
- Make sure the green safety sleeve is stable and is as flat as possible against your skin. If the device is not stable during the injection, you risk bending the needle.
- DO NOT pinch the skin to avoid unintentional needlestick injury.
- Do **NOT** touch or press the blue button while positioning the autoinjector onto your skin.



- Push the open end of the autoinjector against the skin at a 90-degree angle. Apply enough pressure to slide the green safety sleeve up and to maintain it inside the clear cover. Only the wider portion of the green safety sleeve remains outside of the clear cover.
- Do **NOT** press the button until **after** the safety sleeve slides into the clear cover. Pushing the blue button before the safety sleeve is depressed can lead to device failure.
- Inject without pinching the skin.

Press Button to Inject



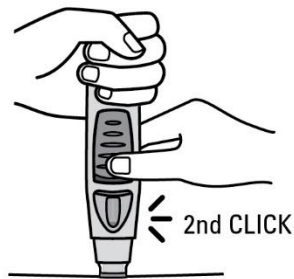
- **Continue to hold the autoinjector against the skin. Use your other hand to press the front raised part of the button** to start your injection. Do not press the button unless the autoinjector is pressed against your skin and the safety sleeve slides into the clear cover.
- Once the button is pressed, it will remain pressed in so you do not need to keep pressure on it.

Tip: If the button seems hard to depress, don't press the button harder. Let go of the button, then lift the autoinjector and start again. Ensure no pressure is on the button until the green safety sleeve is fully depressed against the skin, then press the button.

- **You will hear a loud 'click' sound – don't be alarmed.** The first loud 'click' indicates that the needle has been inserted and the injection has started. You may or may not feel a needle prick at this time.

DO NOT lift the autoinjector away from your skin. If you pull the autoinjector away from the skin, you may not get your full dose of medicine.

Wait for Second 'Click', it usually takes about 3 to 6- seconds, but may take up to 15- seconds for you to hear the second 'Click' sound.



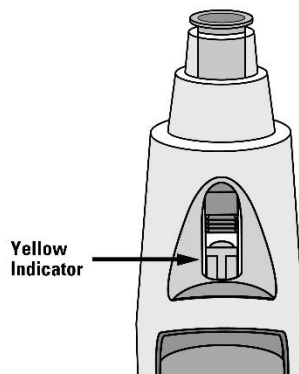
- Continue to hold the autoinjector against the skin until you hear the second 'click' (indicating that the injection has finished and the needle has retracted into the autoinjector).
- Lift the autoinjector from the injection site.

Note: If you do not hear the second 'click', wait 15 seconds from the time you first press the button and then lift the autoinjector from the injection site.

STEP 4: AFTER THE INJECTION

Check the Viewing Window

- After injecting, check the viewing window to make sure that the yellow indicator is visible.
- The yellow indicator may not fill the entire viewing window. This is normal.
- This indicates that the autoinjector has worked properly.
- If you do not think you received your injection, check the yellow indicator again to confirm that the dose was delivered.



- If the yellow indicator is not visible in the viewing window, call your doctor or pharmacist for assistance. Do **NOT** administer a second dose without speaking to your doctor.

Disposing of the Autoinjector

- Immediately dispose of the autoinjector in the sharps container.



- Dispose of the sharps container according to your local regulations when the container is full.

Use Cotton Ball or Gauze

- There may be a small amount of blood or liquid at the injection site, which is normal.
- You can press a cotton ball or gauze over the injection site for 10 seconds.

Do **NOT** rub the injection site.

- You may cover the injection site with a small adhesive bandage, if necessary.

MANUFACTURER

SIMPONI Solution for Injection

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CILAG AG

HOCHSTRASSE 201, 8200 SCHAFFHAUSEN, SWITZERLAND

SIMPONI IV

CILAG AG

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DATE OF REVISION

7 Oct 2025 (based on CCDS v28Aug2025)